AI VIETNAM
All-in-One Course
(TA Session)

Docker

Extra Class: MLOps

Nguyen-Thuan Duong – TA

Year 2024

Outline

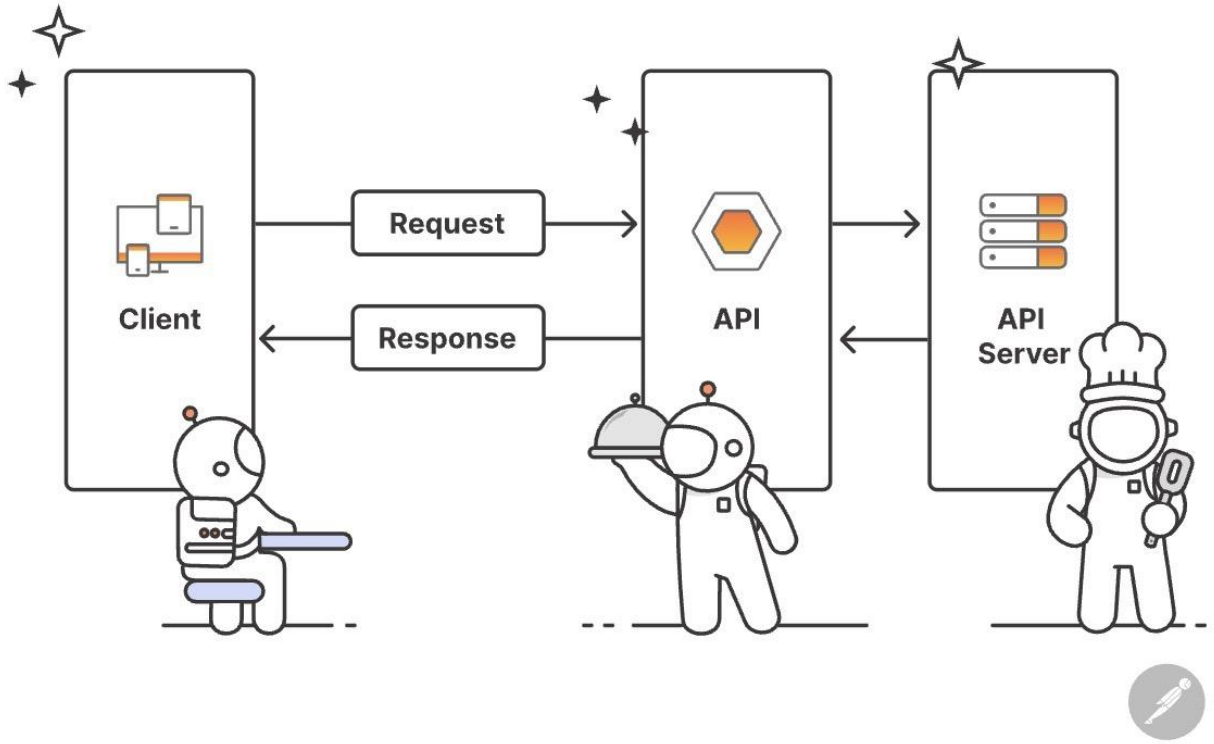
- Introduction
- Docker
- Docker Build
- Docker Run
- Docker Compose
- Docker Hub
- Question

Introduction

Introduction

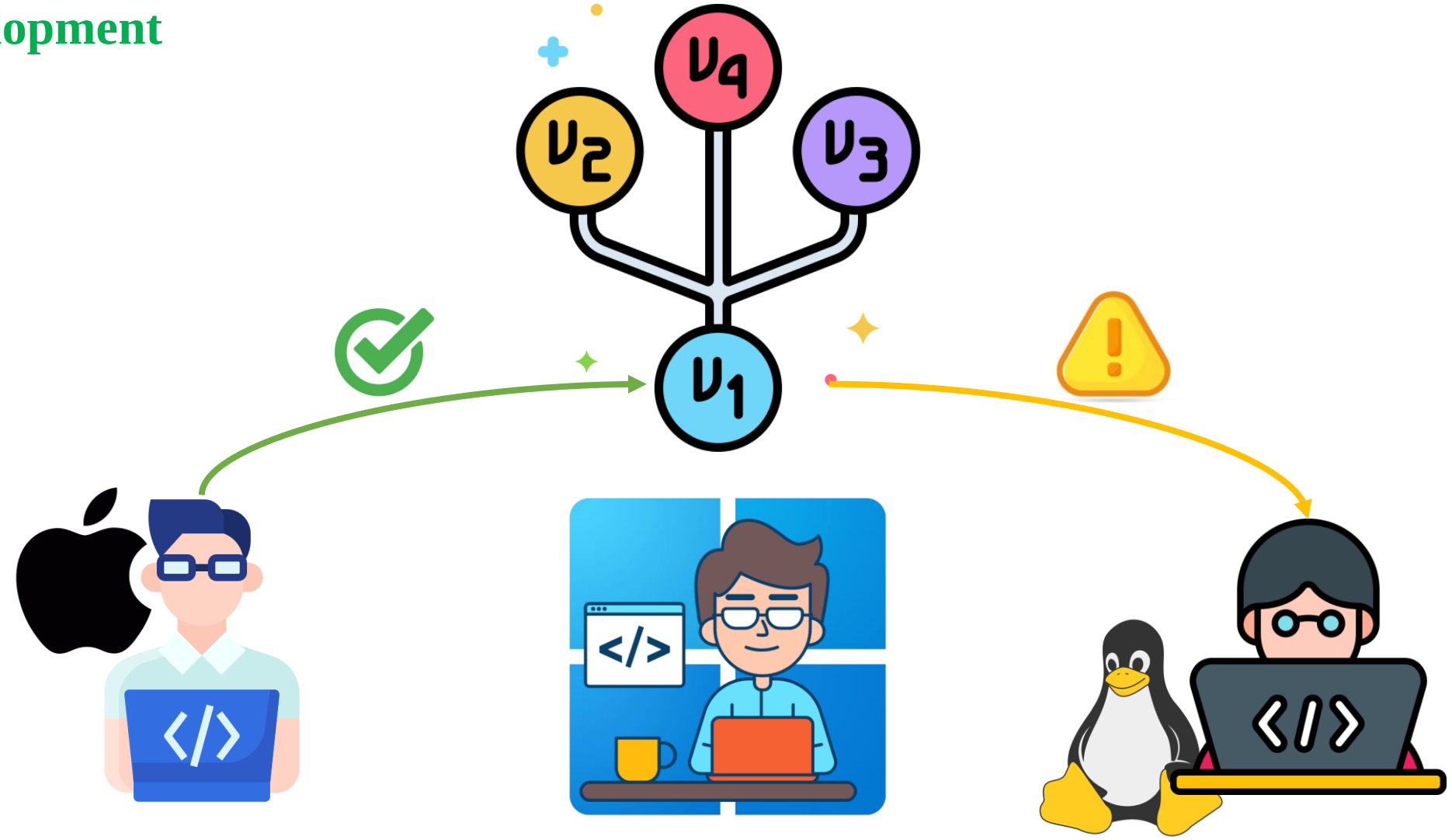
❖ Getting Started

- Frontend (User Interface);
  - HTML, CSS, JS
  - React, Flutter, …
  - Streamlit UI, Gradio UI, …
- Backend:
  - Server: NodeJS, FastAPI
  - APIs: REST Full, GraphQL
  - Database: SQL, NoSQL
- Infrastructure:
  - Hosting: GC, AWS
  - Networking
  - Containers
  - Monitoring



Introduction

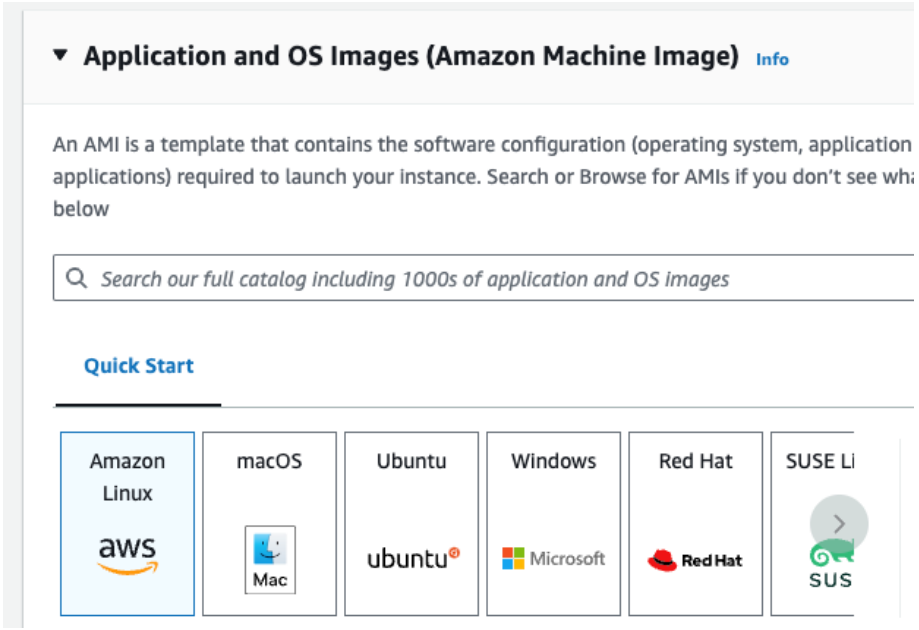
❖ Development



Introduction

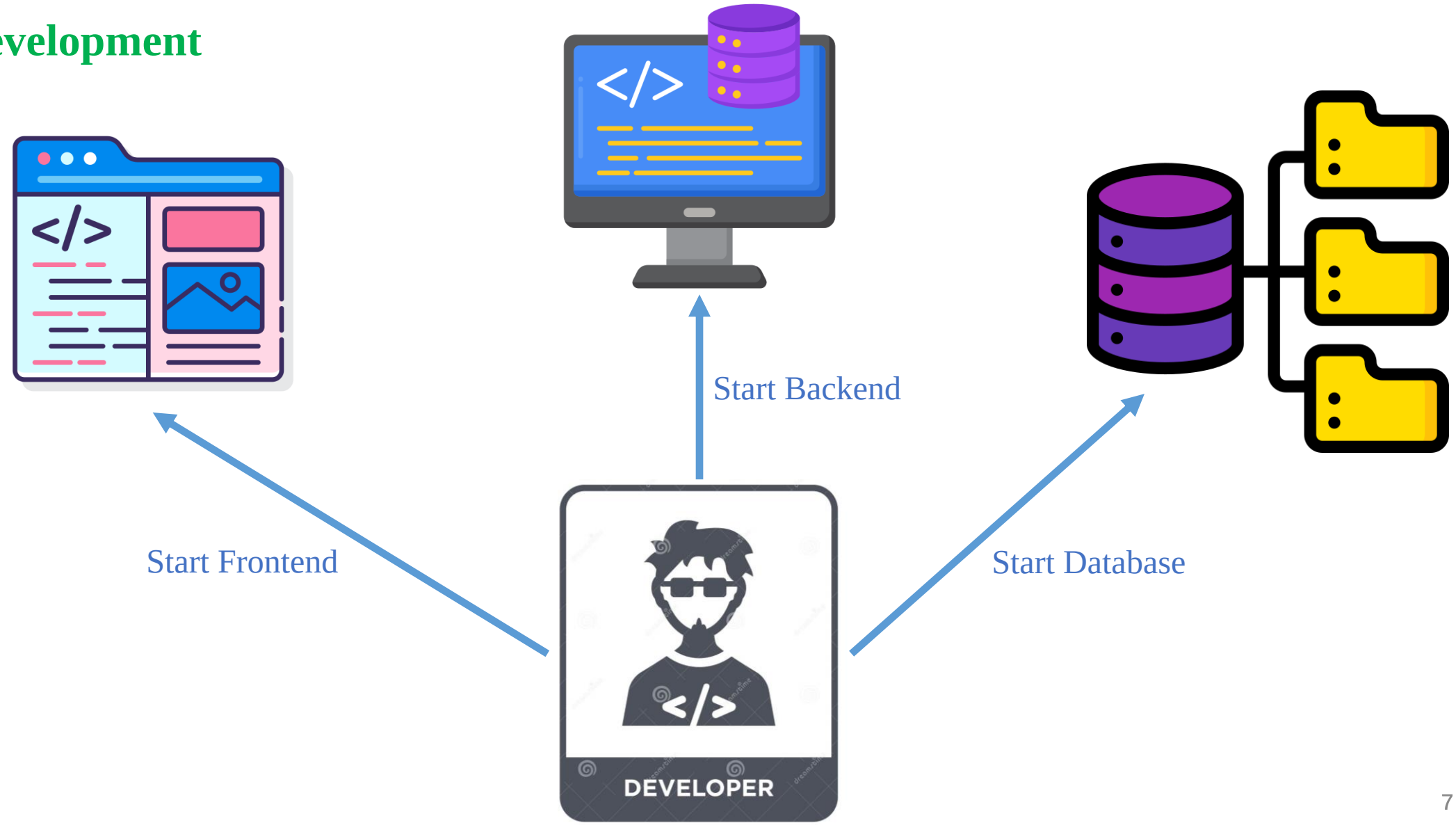
❖ Deployment

Your Application



Introduction

❖ Development



Introduction

❖ Deployment

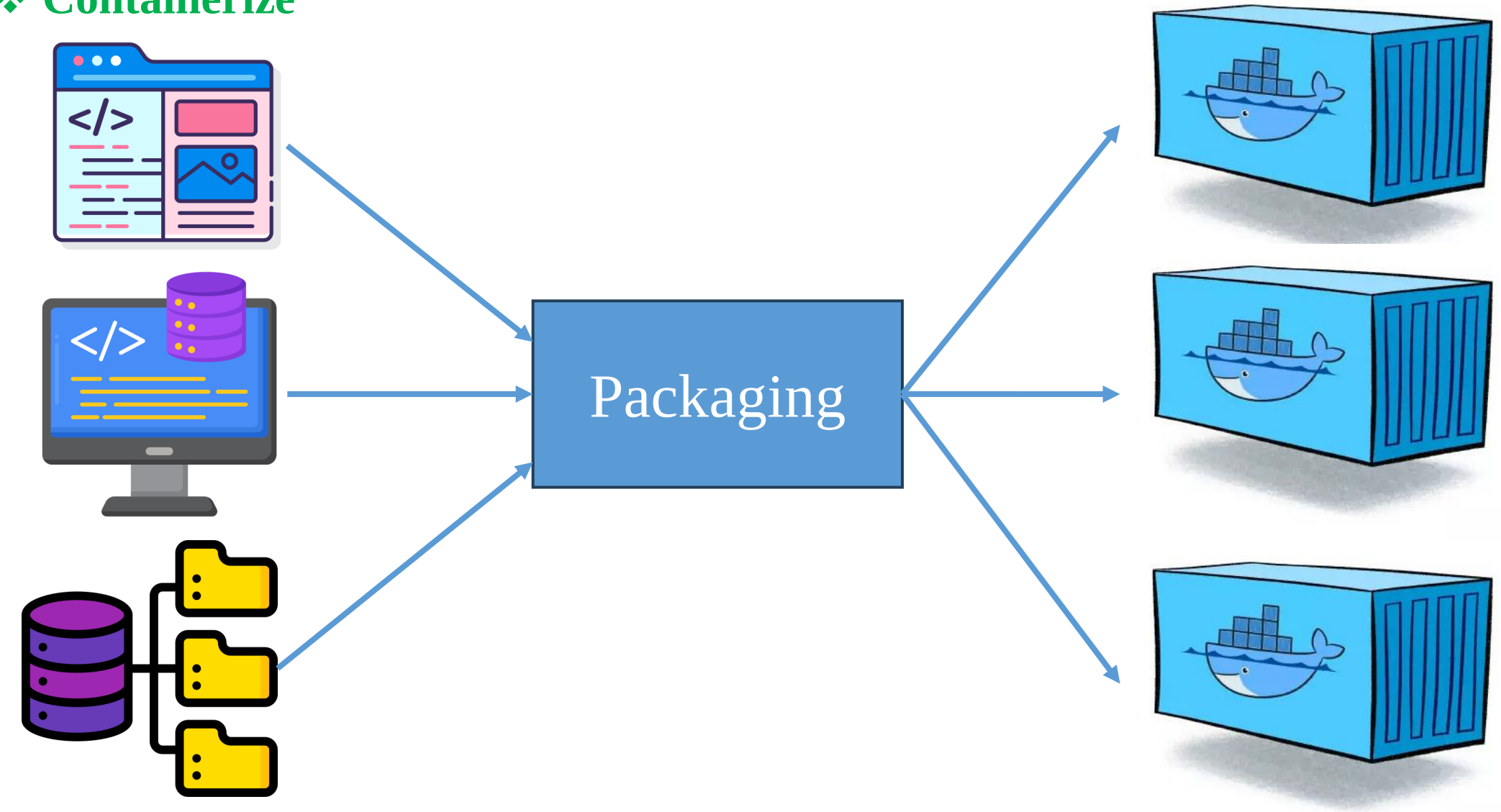
Config/Setup
Backend Server

Config/Setup
Frontend Server

Config/Setup
Database Server

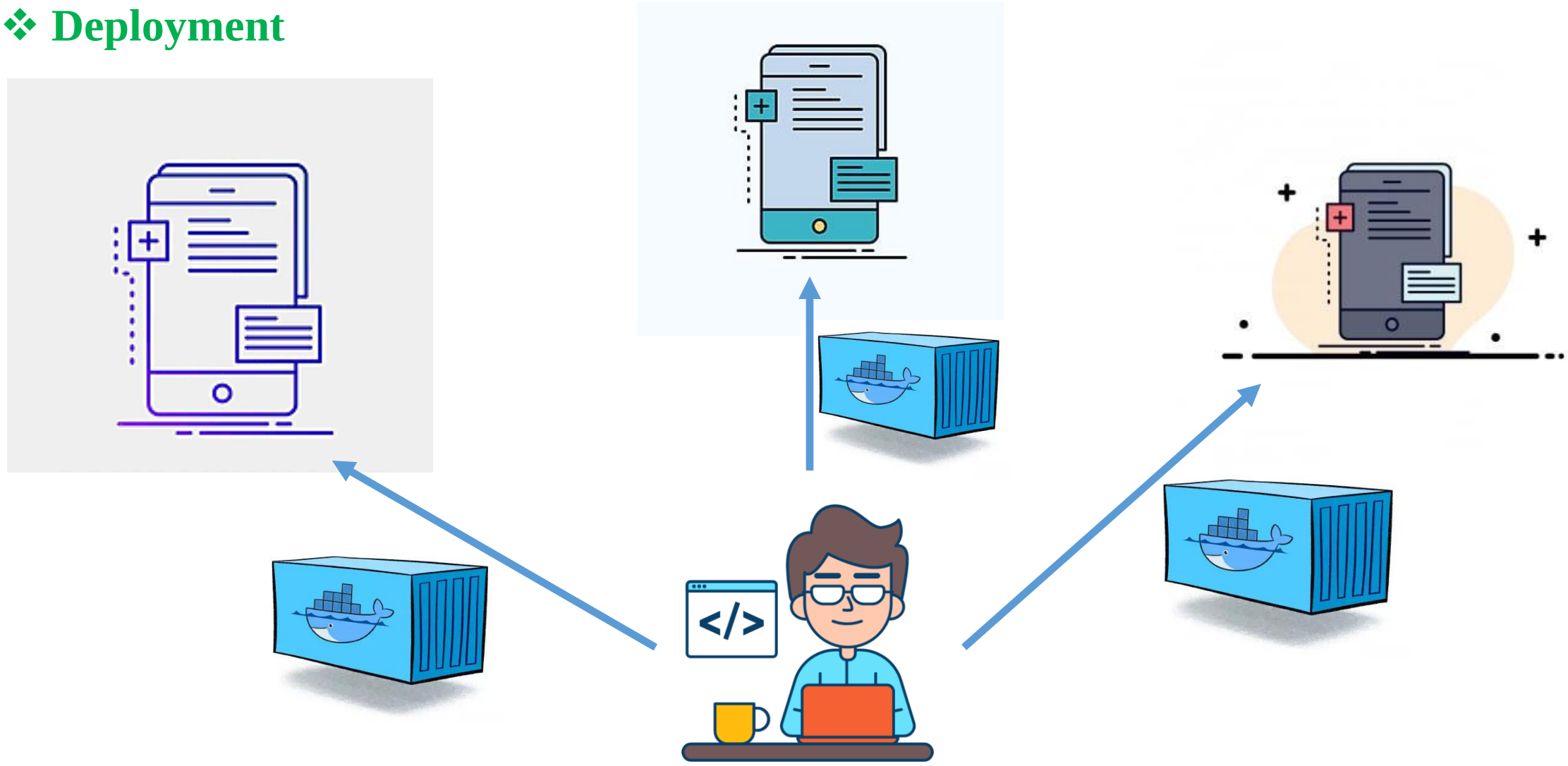
Introduction

❖ Containerize



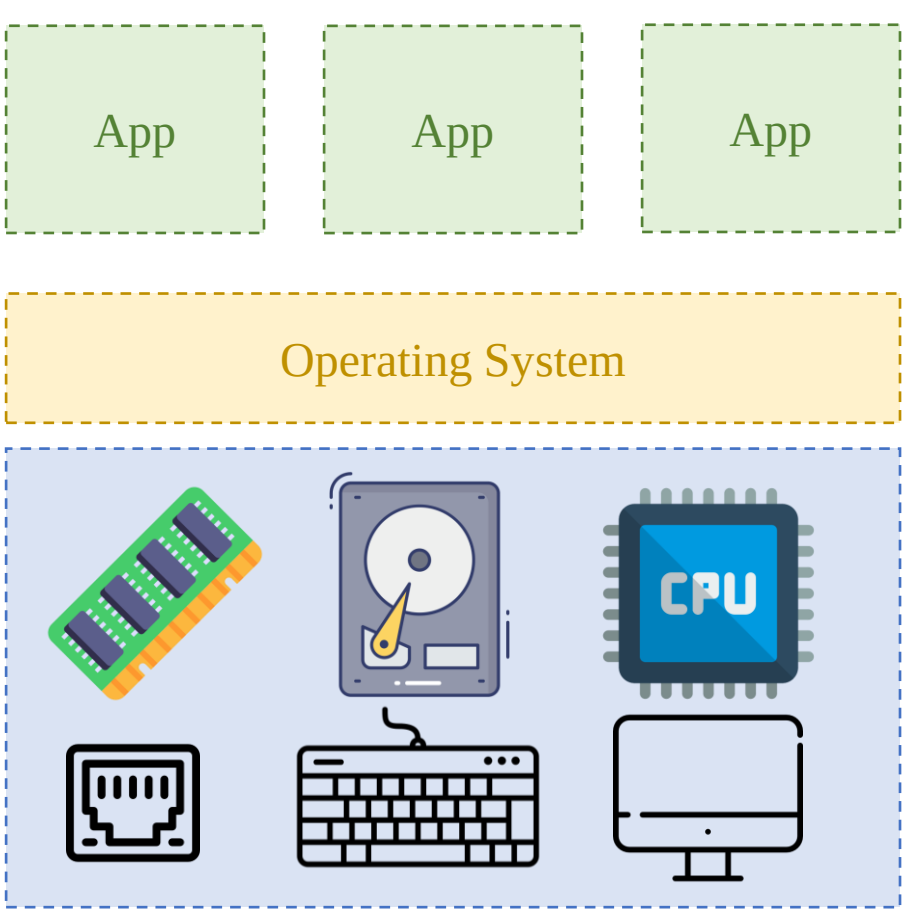
Introduction

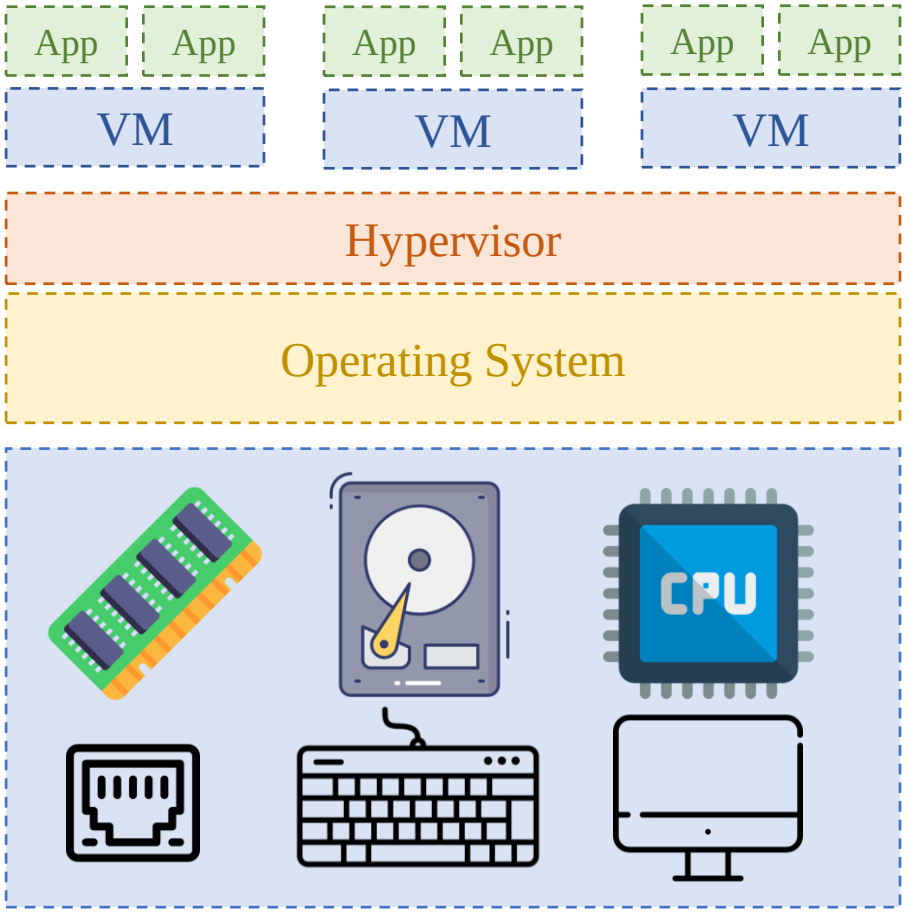
❖ Deployment



Introduction

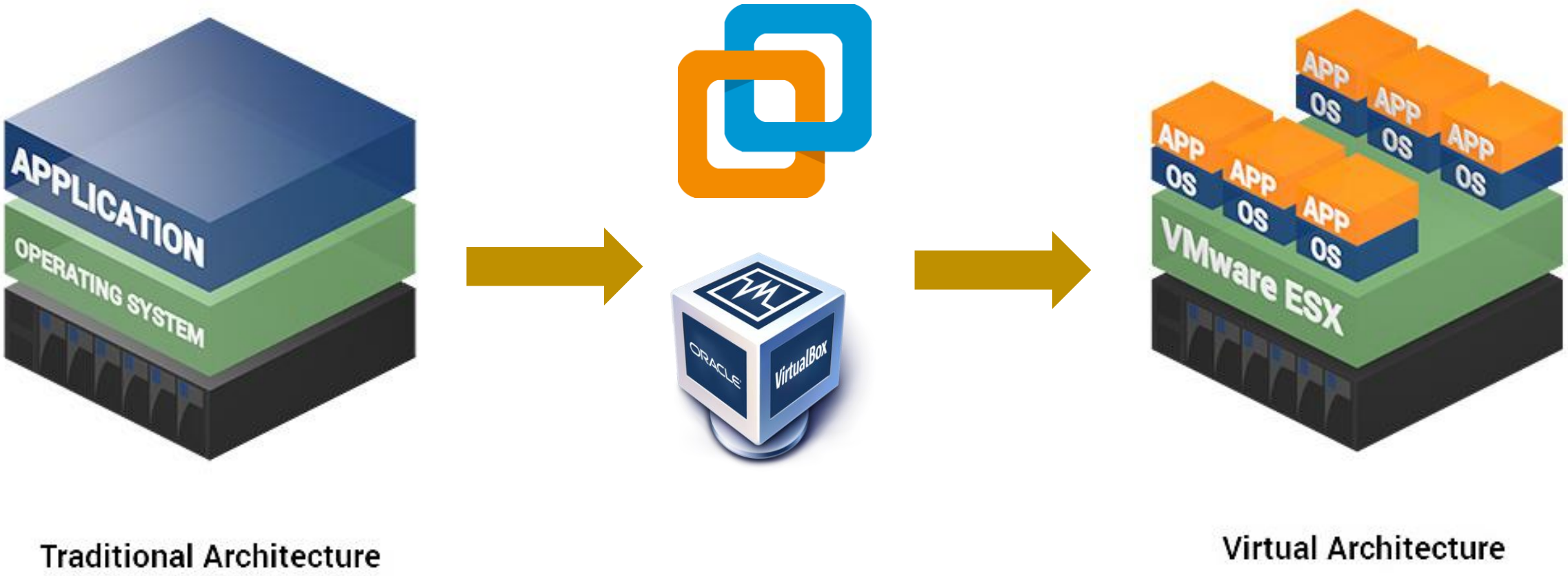
❖ Previous Solution





Introduction

❖ Previous Solution

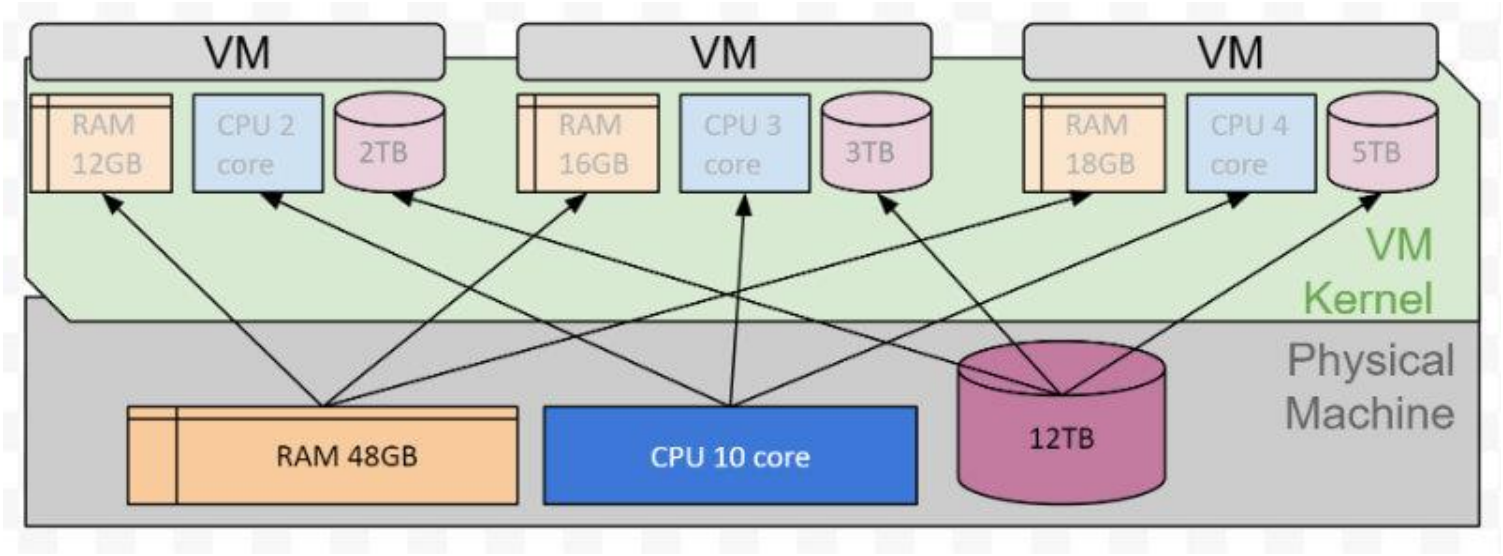


Traditional Architecture

Virtual Architecture

Introduction

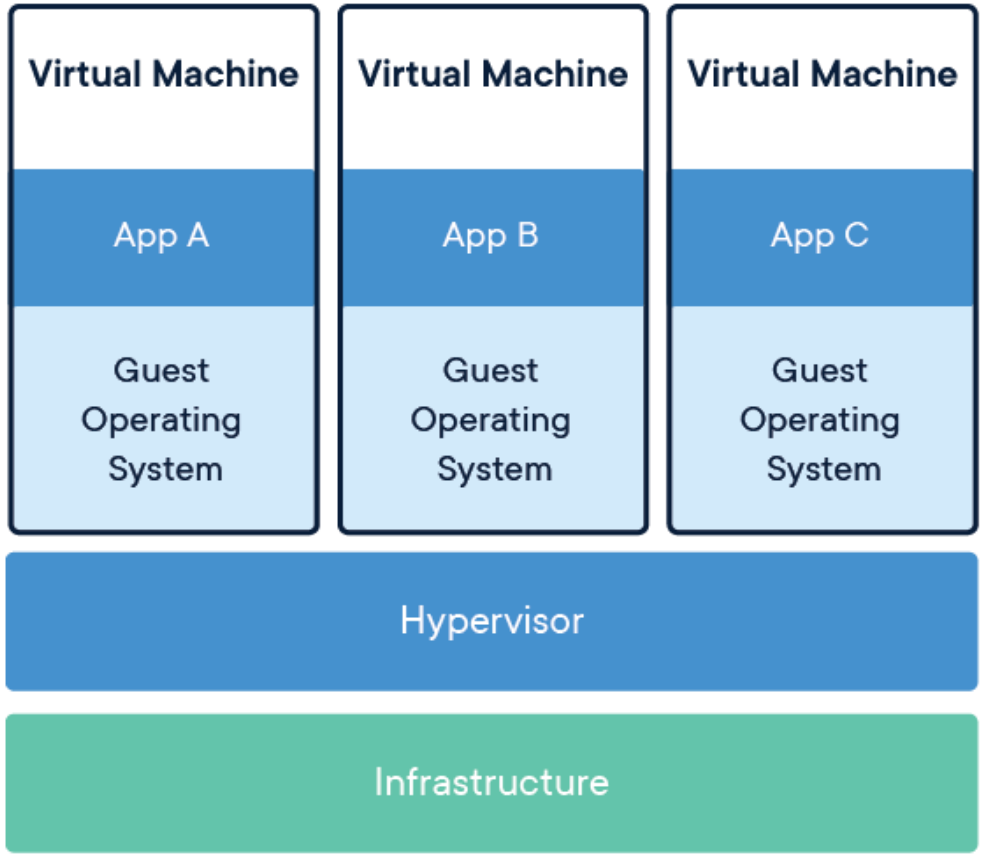
❖ Previous Solution

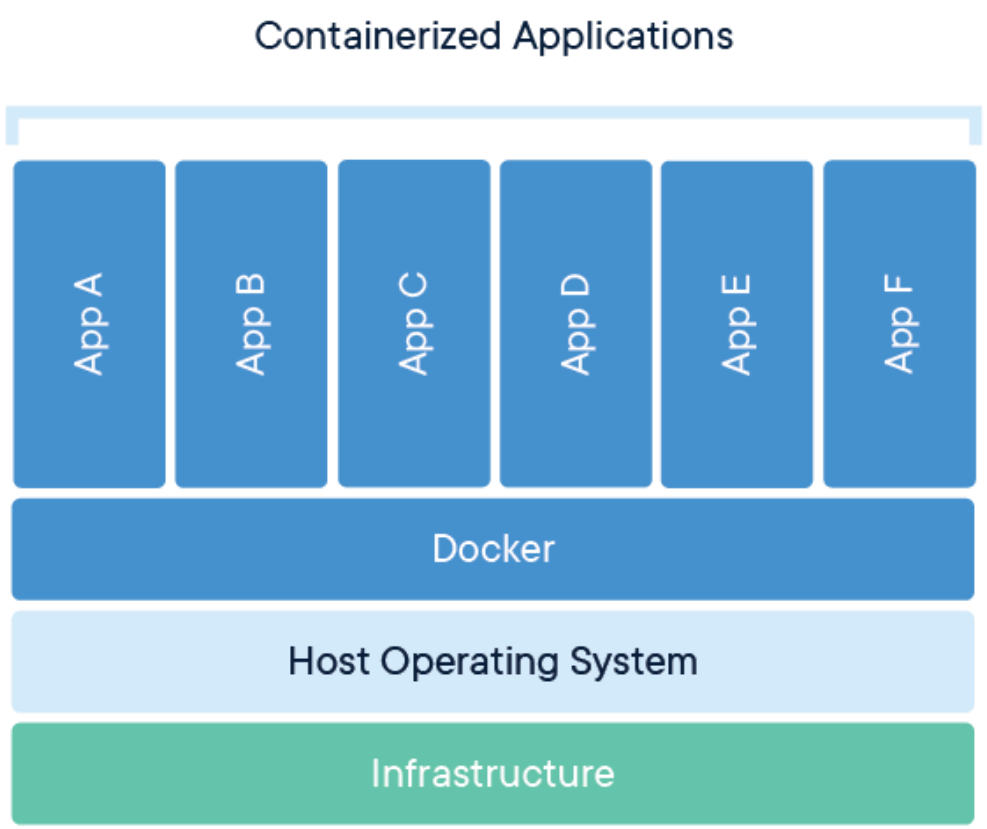


Allocate resources when initializing VM

Introduction

❖ Container Method





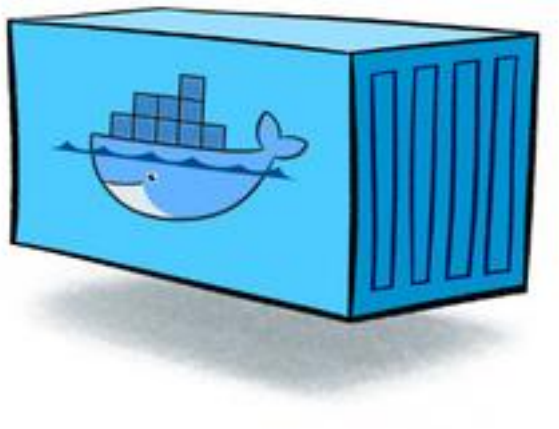
Ref: https://www.docker.com/resources/what-container

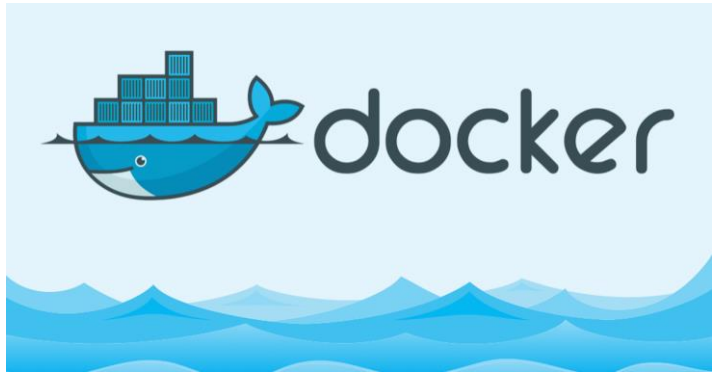
Docker

Docker

❖ Docker

- Docker is an open platform for developing, shipping, and running applications.
- Significantly reduce the delay between writing code and running it in production.

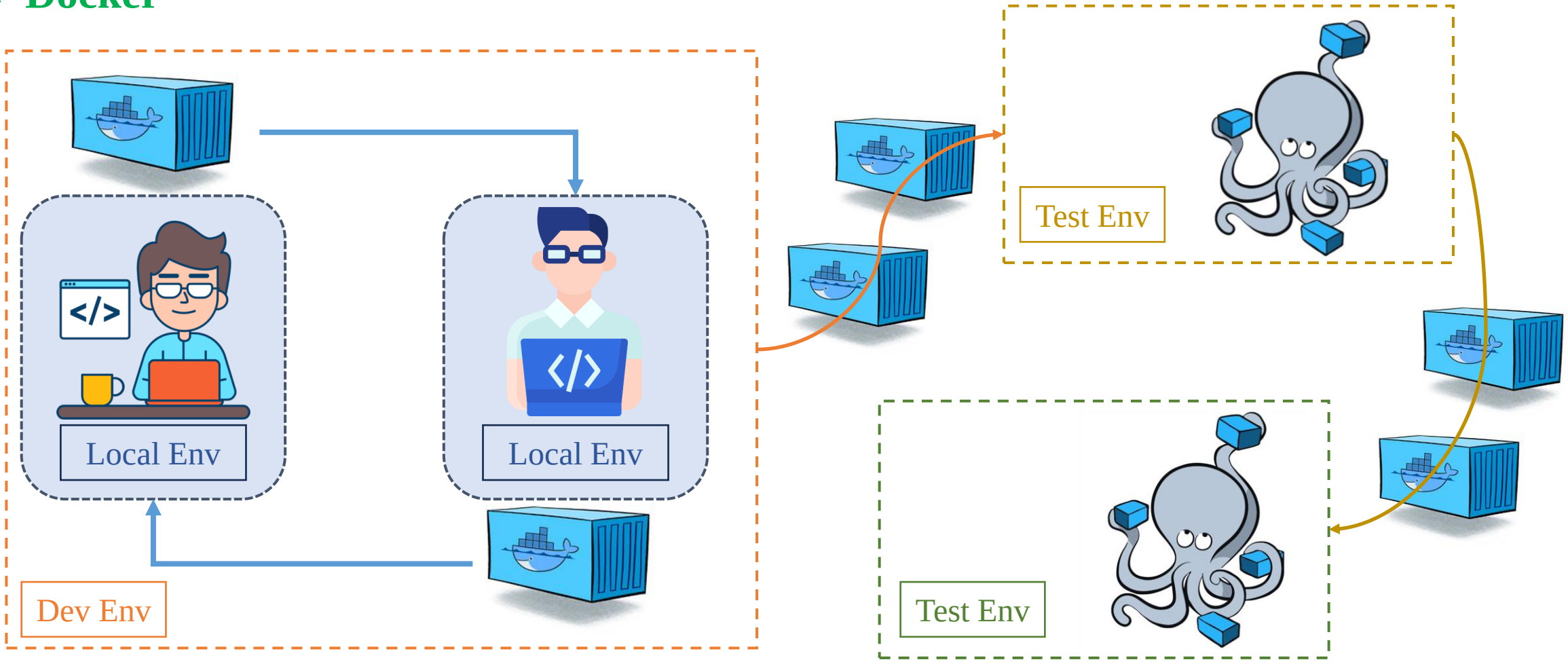




Ref: https://docs.docker.com/get-started/docker-overview

Docker

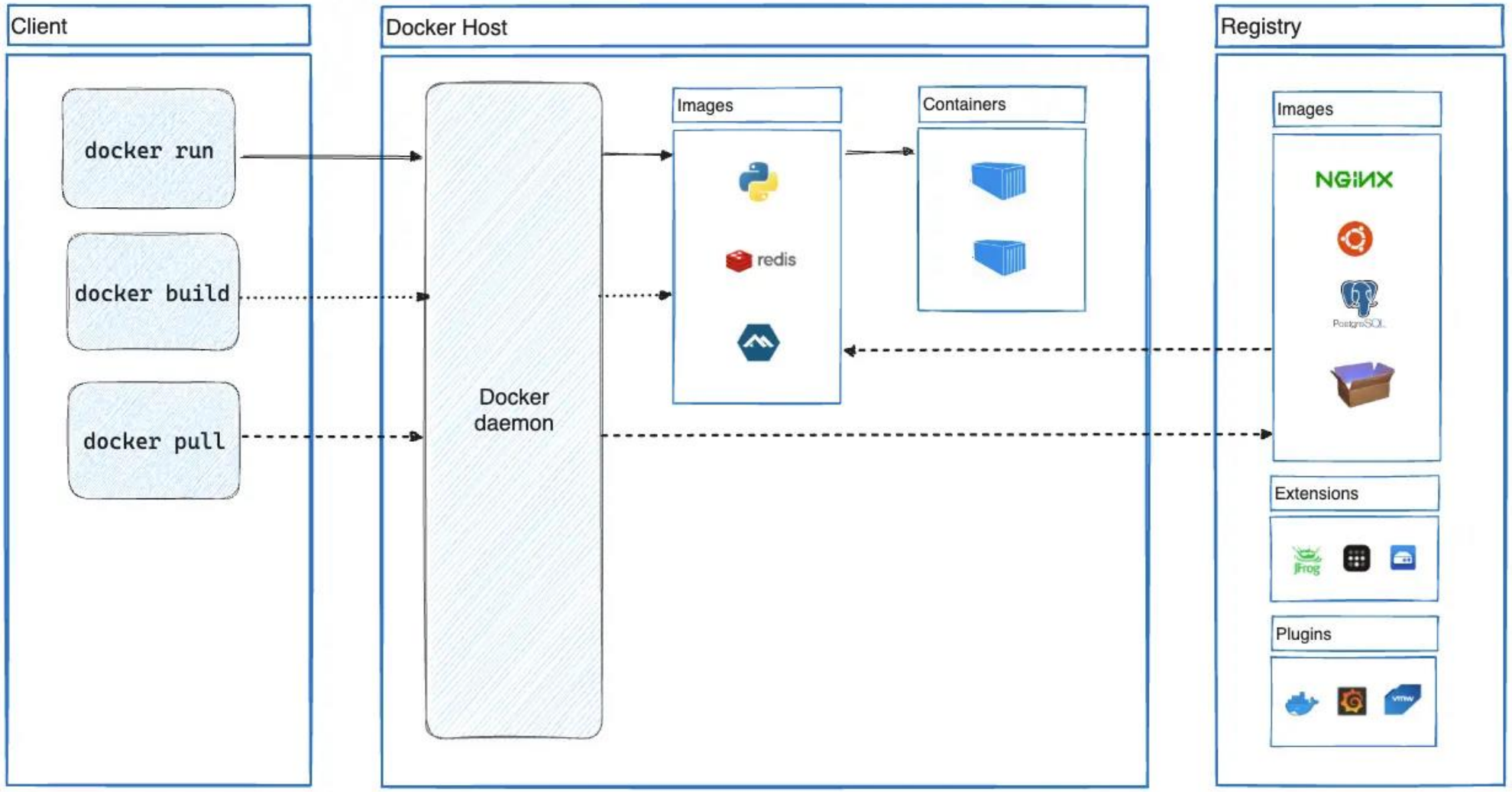
❖ Docker



Ref: https://docs.docker.com/get-started/docker-overview

Docker

❖ Docker Architecture



Ref: https://docs.docker.com/get-started/docker-overview

Docker

❖ Installation

Docker Desktop is a one-click-install application for your Mac, Linux, or Windows environment that lets you build, share, and run containerized applications

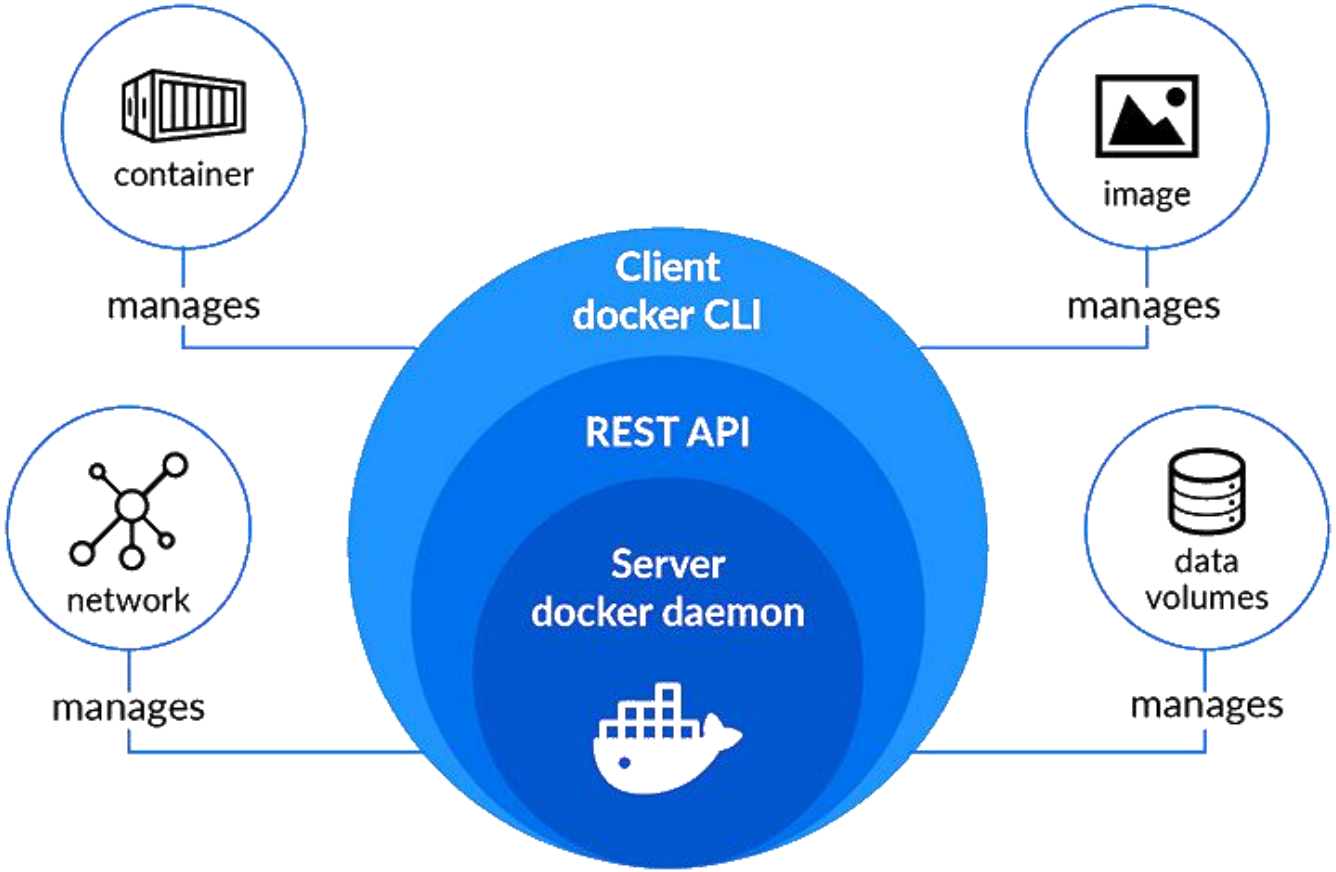
Docker CLI | Docker Engine | GUI

Ref: https://docs.docker.com/desktop

Docker

❖ Deamon

- Listen Docker API requests
- Manages Docker objects such as images, containers, networks, and volumes

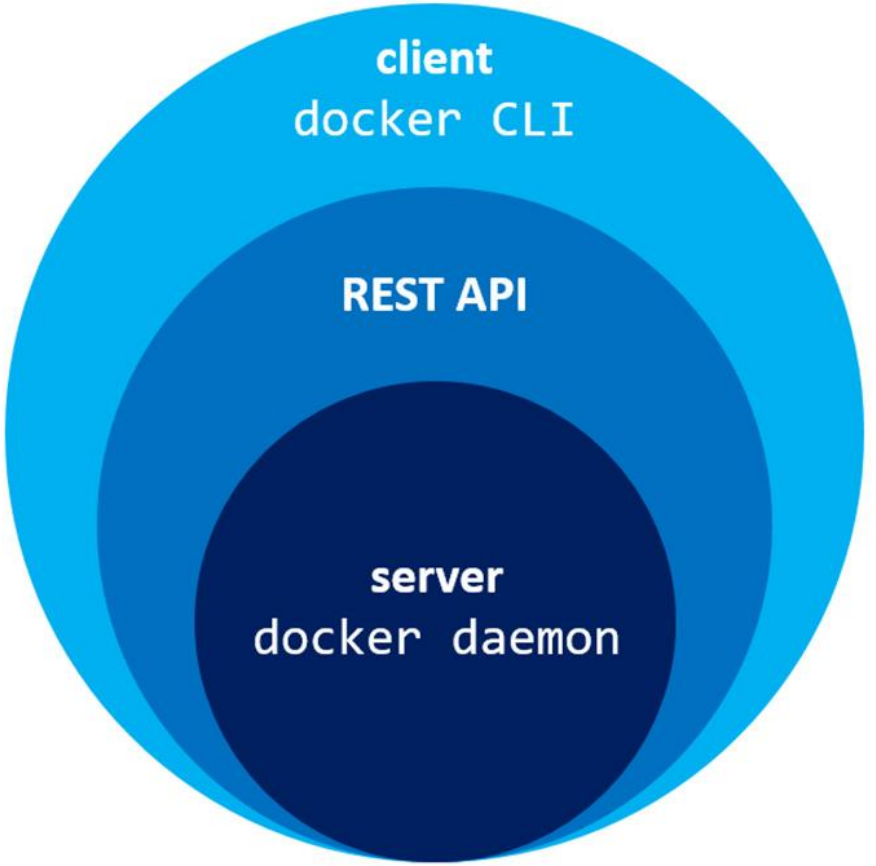


Ref: https://docs.docker.com/get-started/docker-overview

Docker

❖ Engine

- Docker Engine is an open source containerization technology.
- Docker Engine acts as a client-server application.



Ref: https://docs.docker.com/get-started/docker-overview

Docker Engine

❖ Install Docker Engine

Home / Manuals / Docker Engine / Install

Install Docker Engine

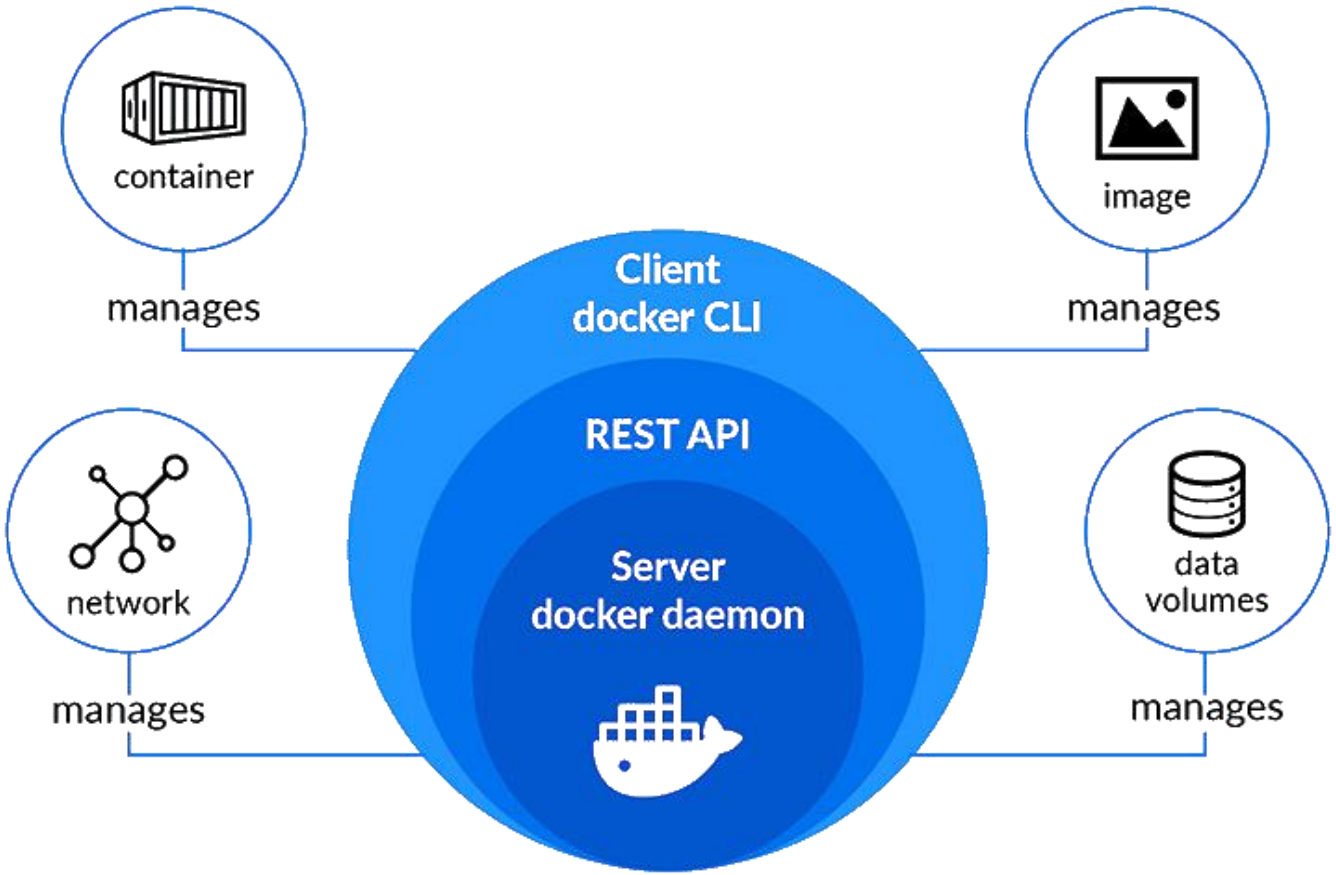
This section describes how to install Docker Engine on Linux, also known as Docker CE. Docker Engine is also ava
Windows, macOS, and Linux, through Docker Desktop. For instructions on how to install Docker Desktop, see:

- Docker Desktop for Linux
- Docker Desktop for Mac (macOS)
- Docker Desktop for Windows

Supported platforms

Platform	x86_64 / amd64	arm64 / aarch64	arm (32-bit)	ppc64le	s390x
CentOS	✅	✅		✅	
Debian	✅	✅	✅	✅	
Fedora	✅	✅		✅	
Raspberry Pi OS (32-bit)			✅		
RHEL	🚧	🚧			✅
SLES					✅
Ubuntu	✅	✅	✅	✅	✅
Binaries	✅	✅	✅		

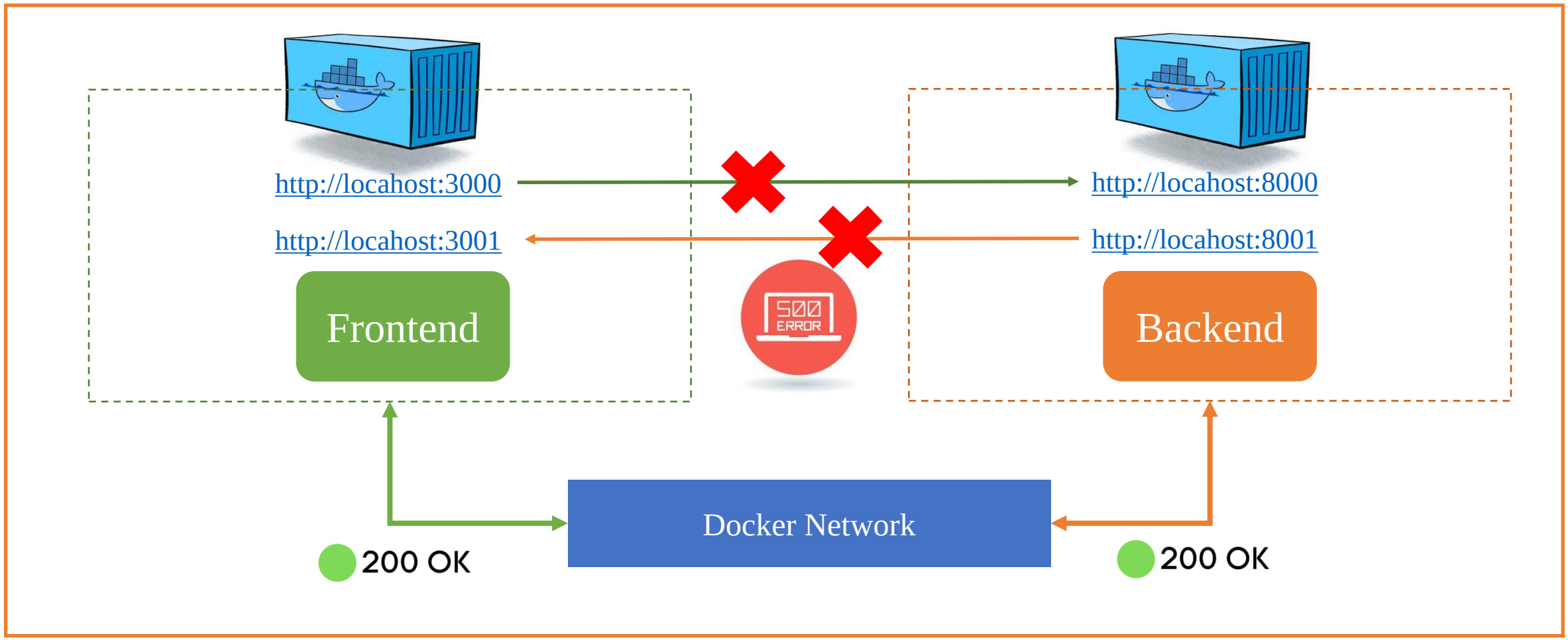
🚧 = Experimental



Ref: https://docs.docker.com/engine/install

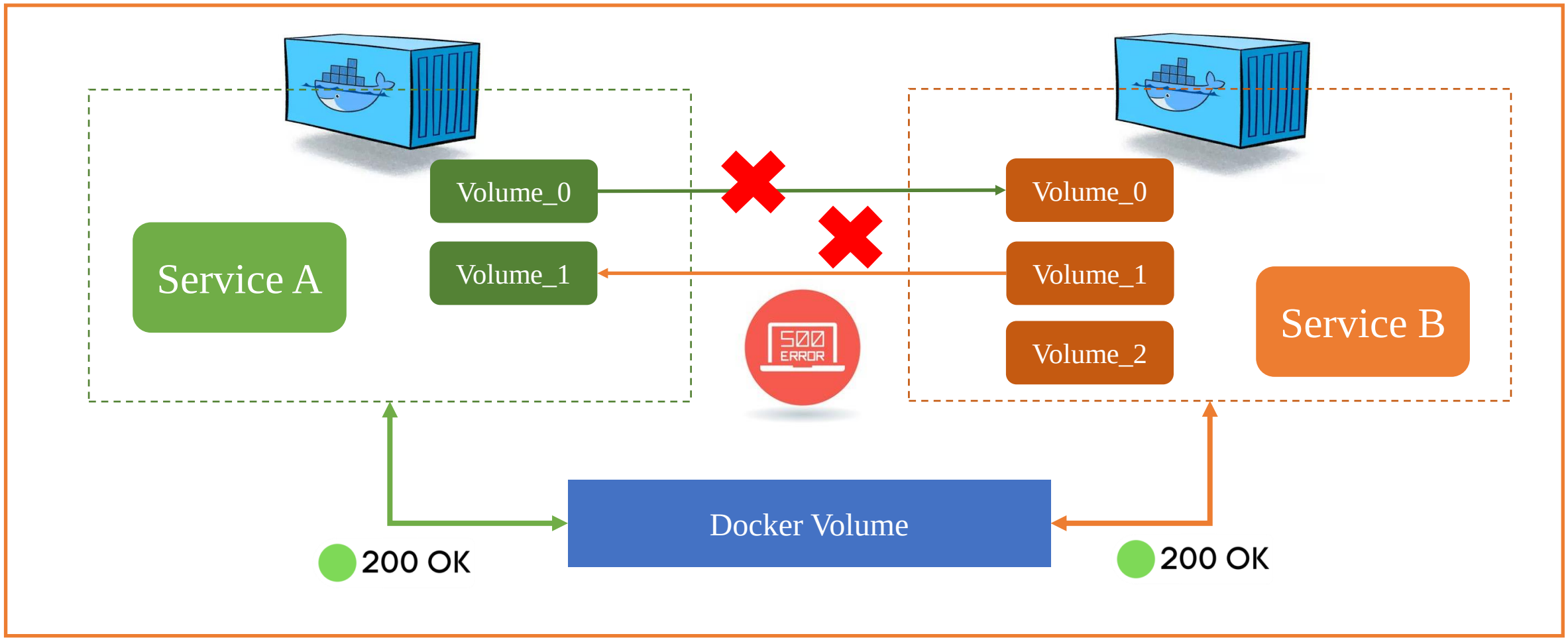
Docker Engine

❖ Networking



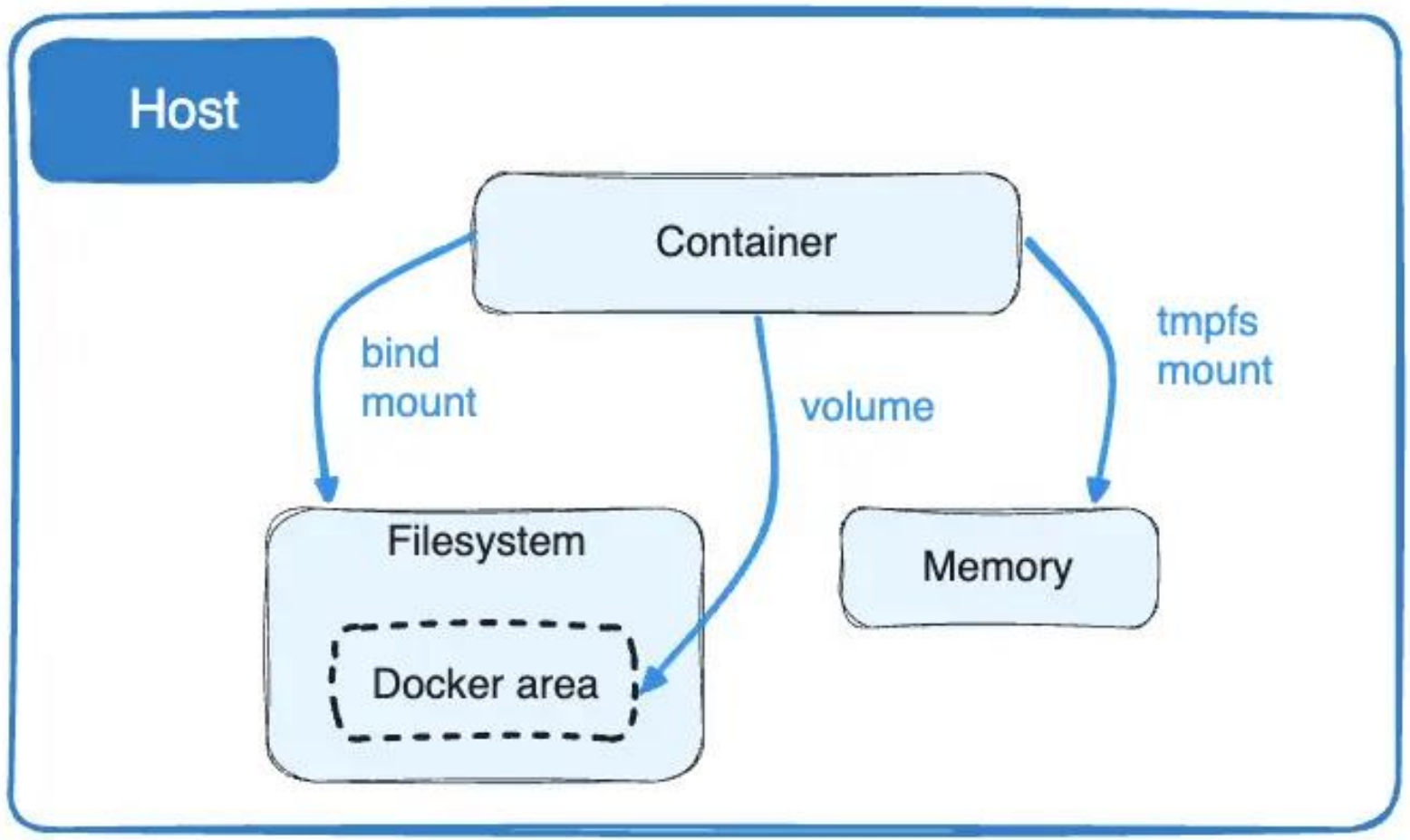
Docker Engine

❖ Storage



Docker Engine

❖ Volumes

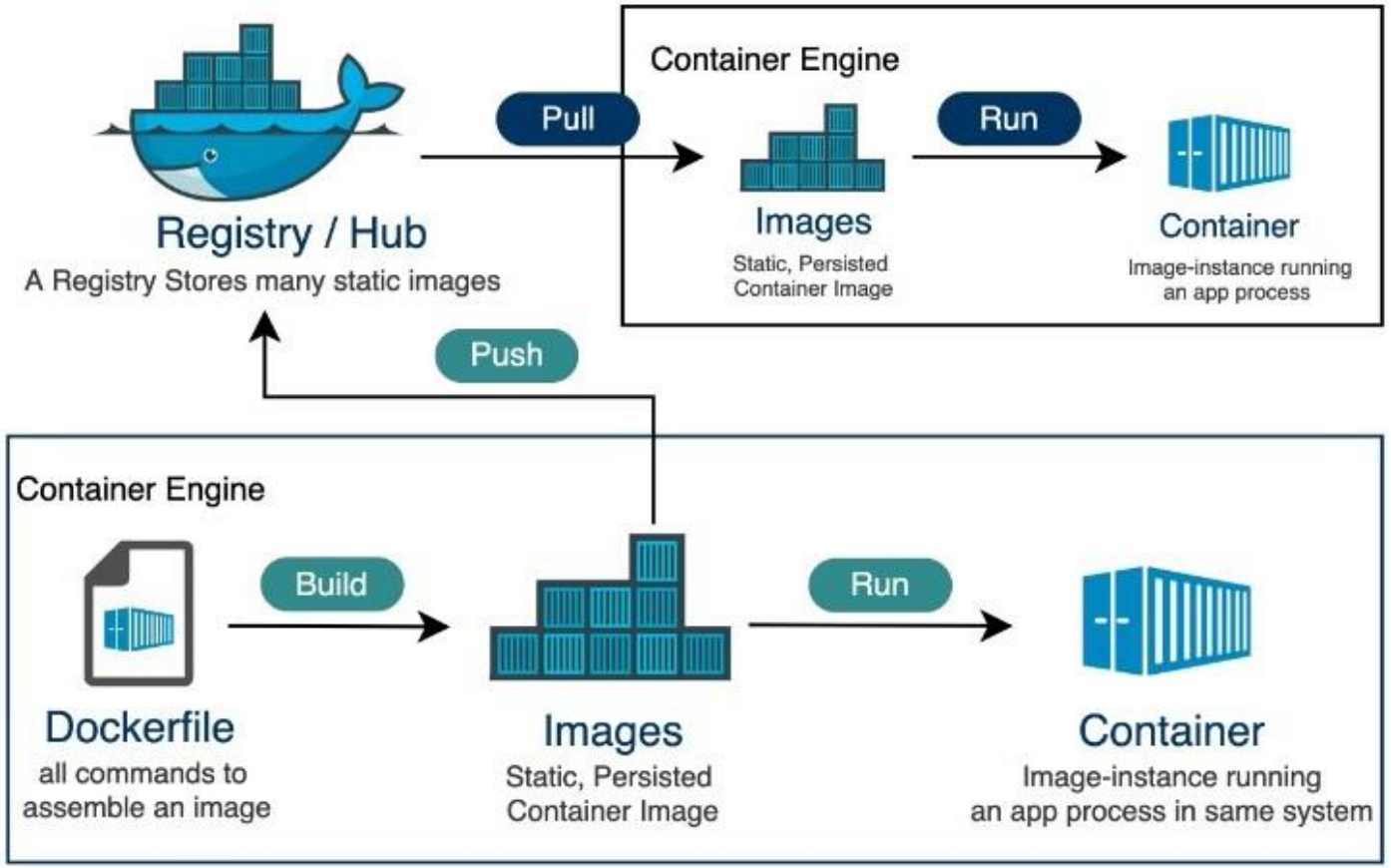


Ref: https://docs.docker.com/engine/storage

Docker

❖ Registry

- A Docker registry stores Docker images.
- Docker Hub is a public registry.

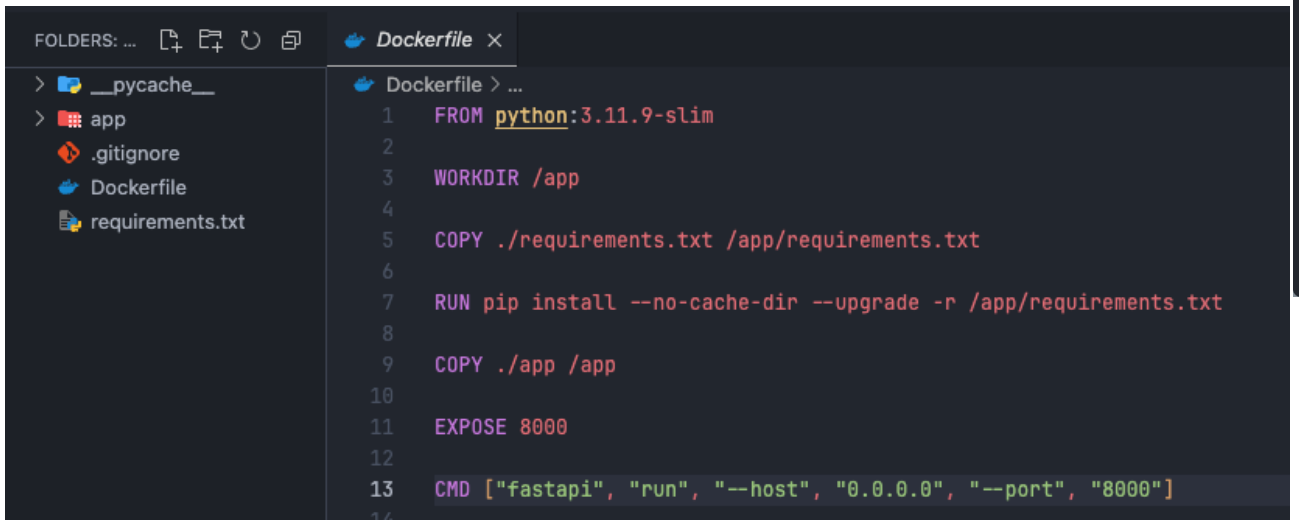


Ref: https://hub.docker.com

Docker

❖ Dockerfile

- A Dockerfile is an instruction to build a a Docker image.
- A Dockerfile is a text file and has instruction syntax defined by the specification



Python Image

```
1 FROM ubuntu:22.04
2
3 RUN apt-get update && apt-get install -y python3 python3-pip
4 RUN pip install fastapi uvicorn
5
6 COPY app.py /app.py
7
8 EXPOSE 8000
9 CMD ["fastapi", "run", "--host", "0.0.0.0", "--port", "8000"]
```

Ubuntu Image

Ref: https://docs.docker.com/build/concepts/dockerfile

Docker

❖ Instructions

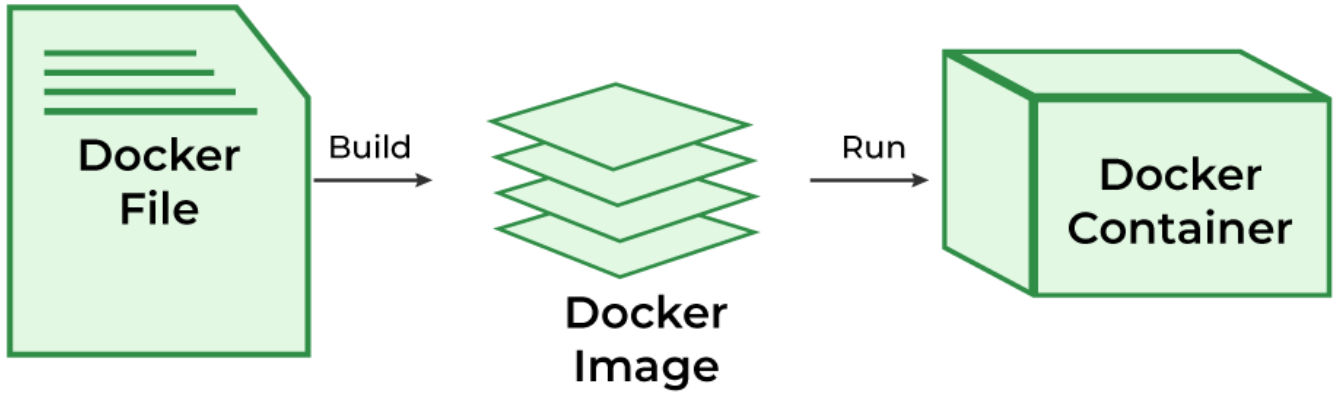
FROM	The base image, which defines the starting point for your build.
RUN	Executes a shell command in Ubuntu
WORKDIR	Set the working directory for any instructions: RUN, CMD, ...
COPY	Copy the file from local into the root directory of image
EXPOSE	Instruction informs Docker that the container listens on the specified network ports at runtime
CMD	Set the command that is run when start the comtainer

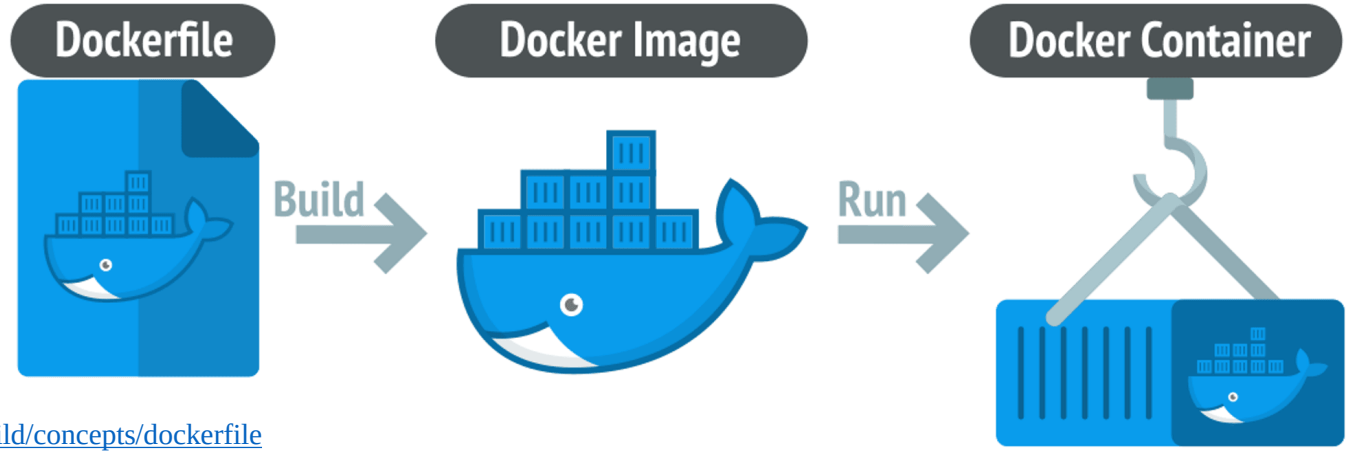
Ref: https://docs.docker.com/reference/dockerfile

Docker

❖ Docker Image

- An image is a template with instructions for creating a Docker container.
- Each instruction in Dockerfile creates a layer in image.

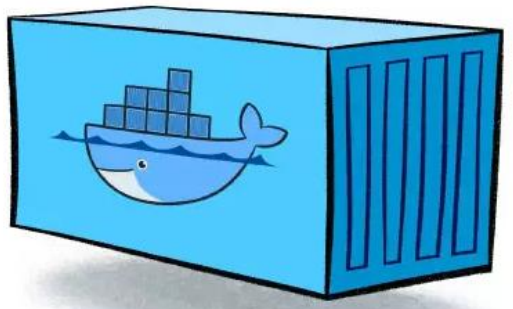




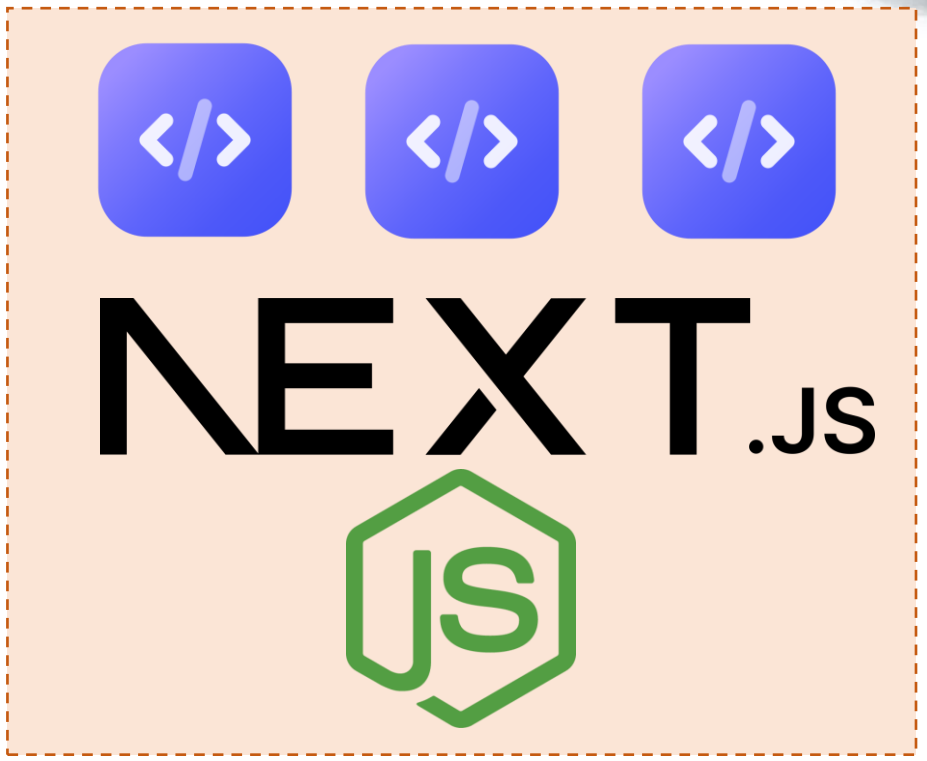
Ref: https://docs.docker.com/build/concepts/dockerfile

Docker

❖ Docker Container



- A container is a lightweight, standalone, and executable software package.
- Include everything an application needs to run: the code, runtime, libraries, and system tools.



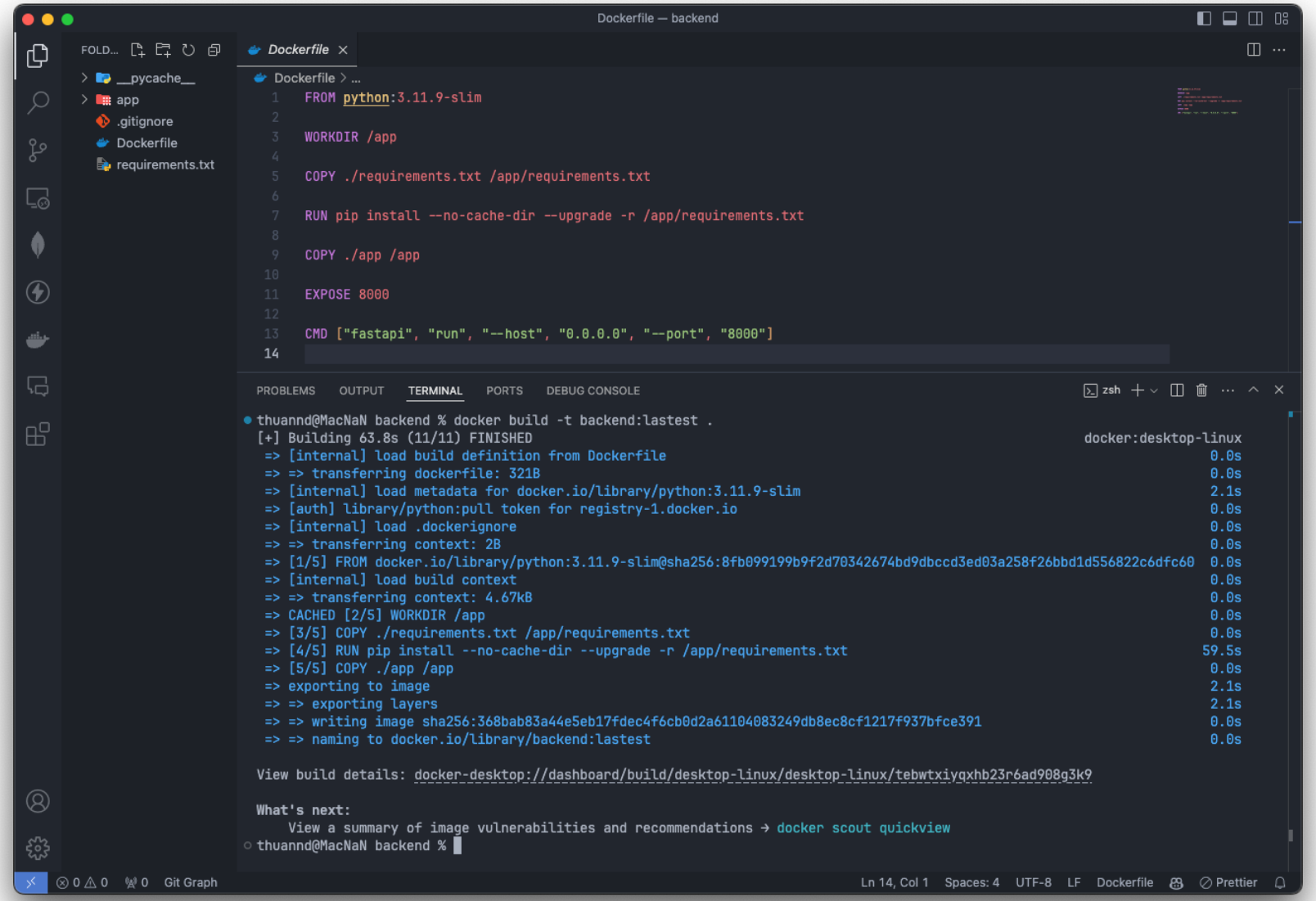
Docker Build

Docker Build

❖ Building

docker build –t <name>:<tag> <build context>

docker build –t backend:lastest .



Docker Build

❖ Building: multi-stage

```dockerfile
FROM python:3.11.9-slim

WORKDIR /app

COPY ./requirements.txt /app/requirements.txt

RUN pip install --no-cache-dir --upgrade -r /app/requirements.txt

COPY ./app /app

EXPOSE 8000

CMD ["fastapi", "run", "--host", "0.0.0.0", "--port", "8000"]
```

```
thuannd@MacNaN backend % docker build -t backend:lastest .
[+] Building 63.8s (11/11) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 321B
 => [internal] load metadata for docker.io/library/python:3.11.9-slim
 => [auth] library/python:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.11.9-slim@sha256:8fb099199b9f2d70342674bd9dbc
 => [internal] load build context
 => => transferring context: 4.67kB
 => CACHED [2/5] WORKDIR /app
 => [3/5] COPY ./requirements.txt /app/requirements.txt
 => [4/5] RUN pip install --no-cache-dir --upgrade -r /app/requirements.txt
 => [5/5] COPY ./app /app
 => exporting to image
 => => exporting layers
 => => writing image sha256:368bab83a44e5eb17fdec4f6cb0d2a61104083249db8ec8cf1217f937bf
 => => naming to docker.io/library/backend:lastest
```

Docker Build

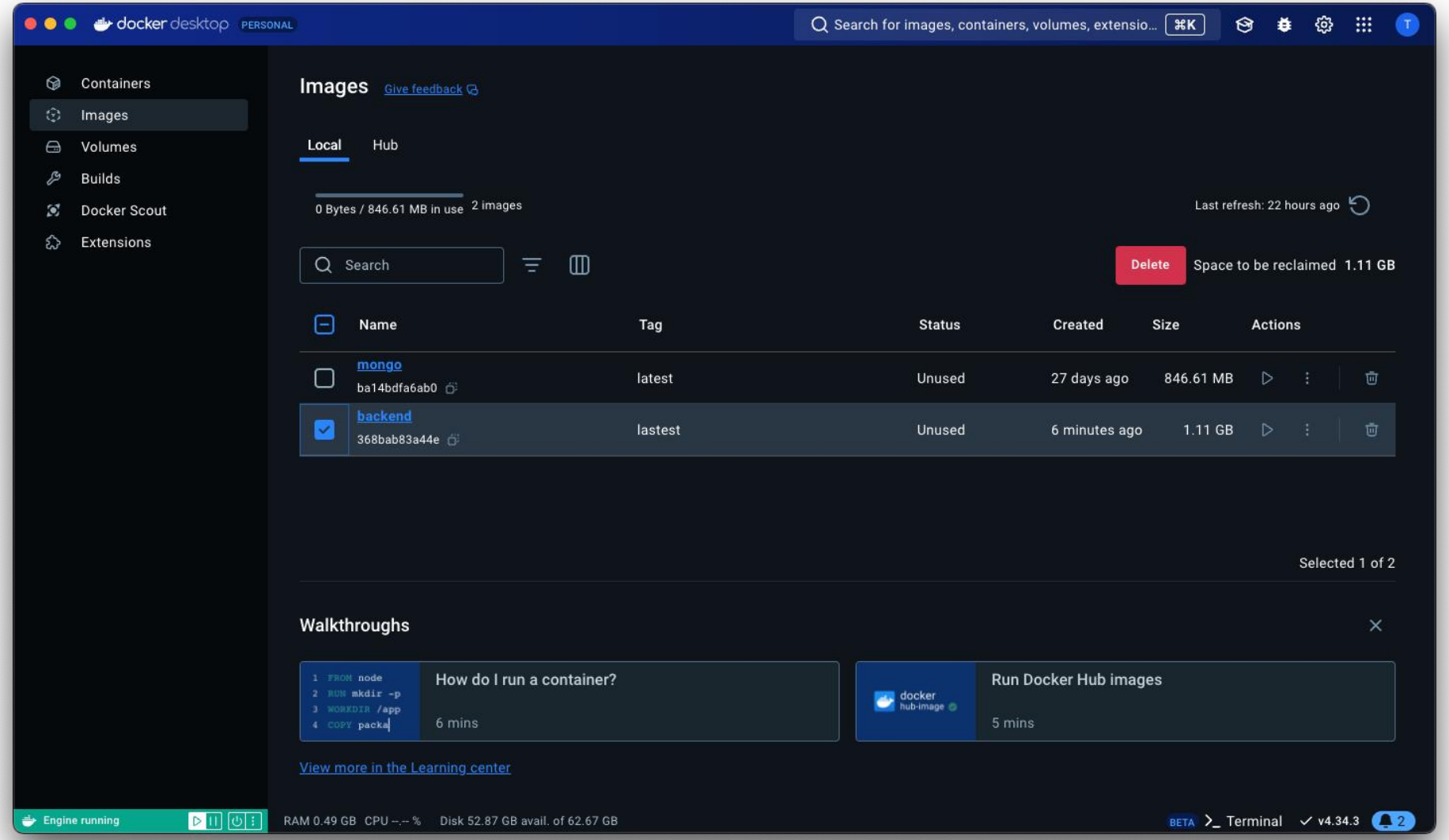
❖ Building: multi-stage

```dockerfile
FROM python:3.11.9-slim

WORKDIR /app

COPY ./requirements.txt /app/requirements.txt

RUN pip install --no-cache-dir --upgrade -r /app/requirements.txt

COPY ./app /app

EXPOSE 8000

CMD ["fastapi", "run", "--host", "0.0.0.0", "--port", "8000"]
```

```
thuannd@MacNaN backend % docker build -t backend:lastest .
[+] Building 63.8s (11/11) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 321B
 => [internal] load metadata for docker.io/library/python:3.11.9-slim
 => [auth] library/python:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.11.9-slim@sha256:8fb099199b9f2d70342674bd9dbc
 => [internal] load build context
 => => transferring context: 4.67kB
 => CACHED [2/5] WORKDIR /app
 => [3/5] COPY ./requirements.txt /app/requirements.txt
 => [4/5] RUN pip install --no-cache-dir --upgrade -r /app/requirements.txt
 => [5/5] COPY ./app /app
 => exporting to image
 => => exporting layers
 => => writing image sha256:368bab83a44e5eb17fdec4f6cb0d2a61104083249db8ec8cf1217f937bf
 => => naming to docker.io/library/backend:lastest
```

Docker Build

❖ Building



Docker Build

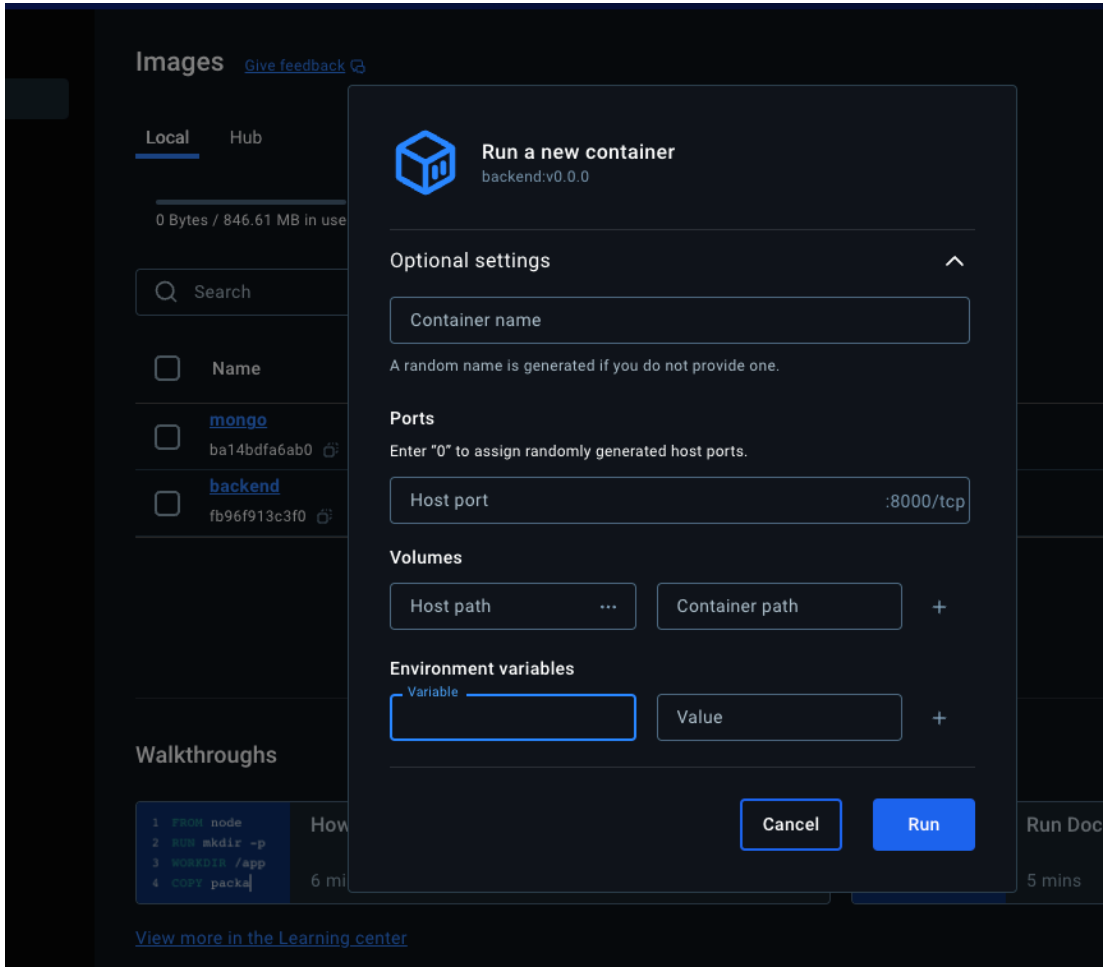
❖ Building: secrets

.env — backend

Dockerfile | .env

.env

```
1  OPENAI_API_KEY=sk_1234567890abcdef1234567890abcdef
```




```
docker build –secret OPENAI_API_KEY=sk_1234567890abcdef1234567890abcdef –t backend:lastest .
```

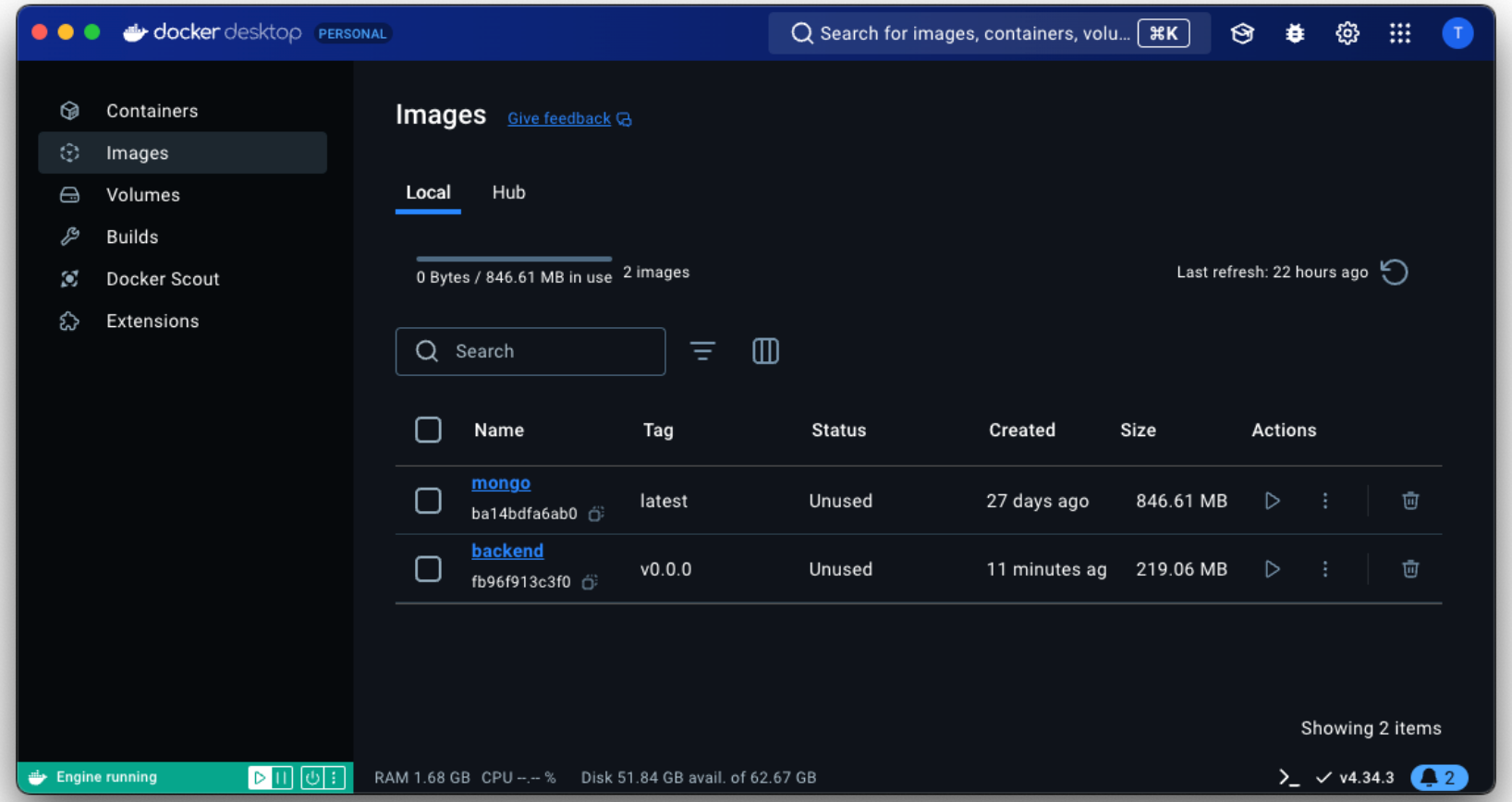
Docker Run

Docker Build

❖ Run container

docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

docker run –it --name backend_cont backend:v0.0.0



Docker Build

❖ Exec to container

docker exec [OPTIONS] CONTAINER COMMAND [ARG...]

```
thuannd@MacNaN backend % docker exec --help

Usage:  docker exec [OPTIONS] CONTAINER COMMAND [ARG...]

Execute a command in a running container

Aliases:
  docker container exec, docker exec

Options:
  -d, --detach               Detached mode: run command in the background
      --detach-keys string   Override the key sequence for detaching a container
  -e, --env list             Set environment variables
      --env-file list        Read in a file of environment variables
  -i, --interactive          Keep STDIN open even if not attached
      --privileged           Give extended privileges to the command
  -t, --tty                  Allocate a pseudo-TTY
  -u, --user string          Username or UID (format: "<name|uid>[:<group|gid>]")
  -w, --workdir string       Working directory inside the container
```

Docker Build

❖ Exec to container

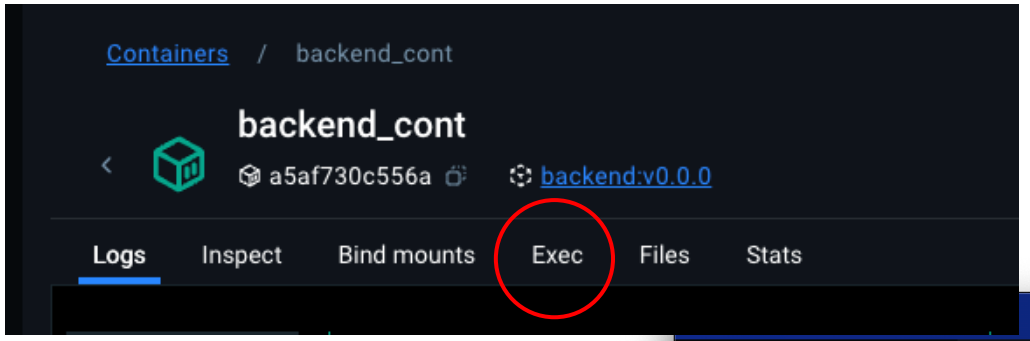
docker exec [OPTIONS] CONTAINER COMMAND [ARG...]

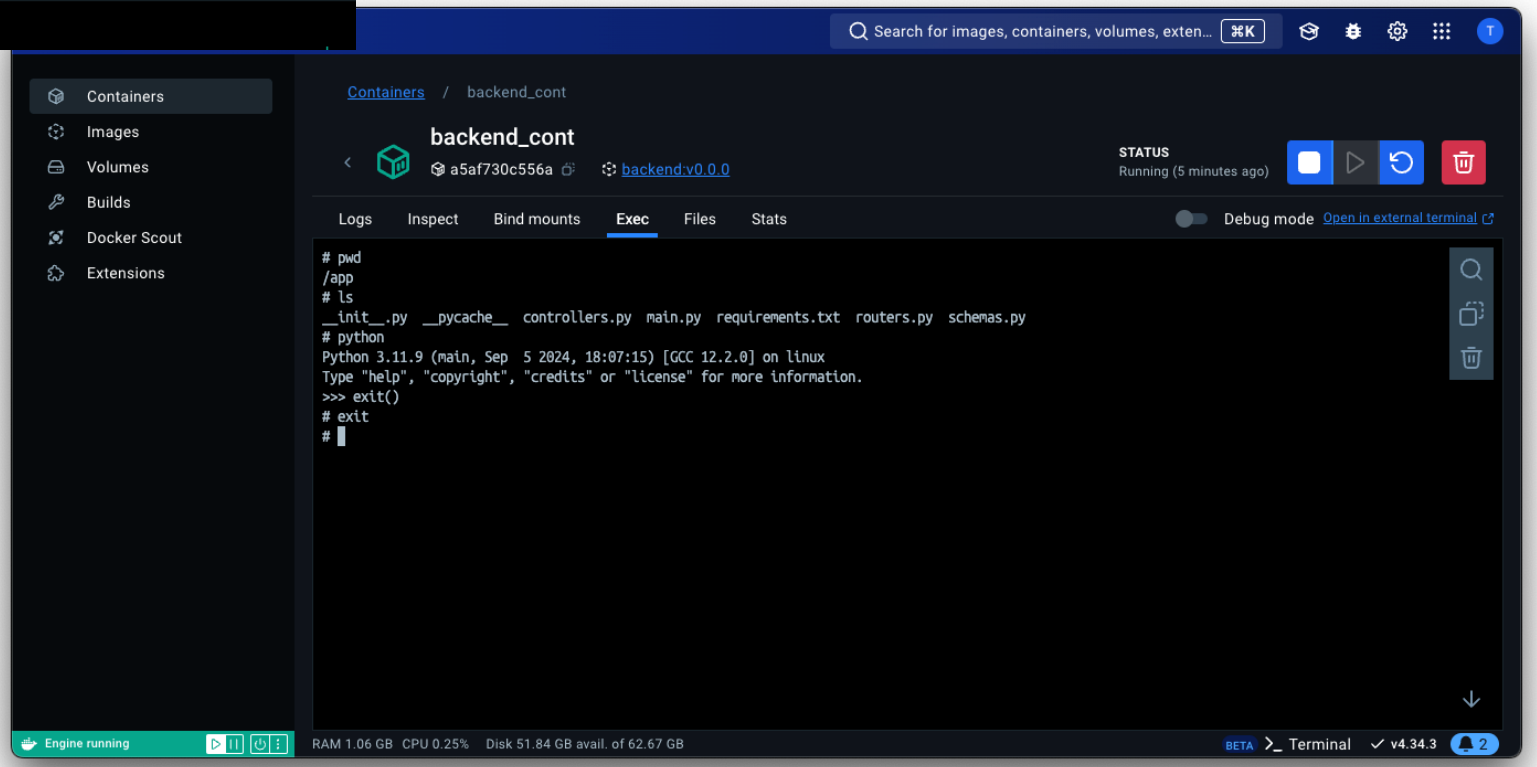
docker exec -it backend_cont bash

```
PROBLEMS  OUTPUT  TERMINAL  PORTS  DEBUG CONSOLE  JUPYTER

thuannd@MacNaN backend % docker exec -it backend_cont bash
root@a5af730c556a:/app# pwd
/app
root@a5af730c556a:/app# ls
__init__.py  __pycache__  controllers.py  main.py  requirements.txt  routers.py  schemas.py
root@a5af730c556a:/app# python
Python 3.11.9 (main, Sep  5 2024, 18:07:15) [GCC 12.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()
root@a5af730c556a:/app# exit
exit
```

Docker Build

❖ Exec to container





Docker Build

❖ Container logs

docker logs [OPTIONS] CONTAINER

docker logs backend_cont

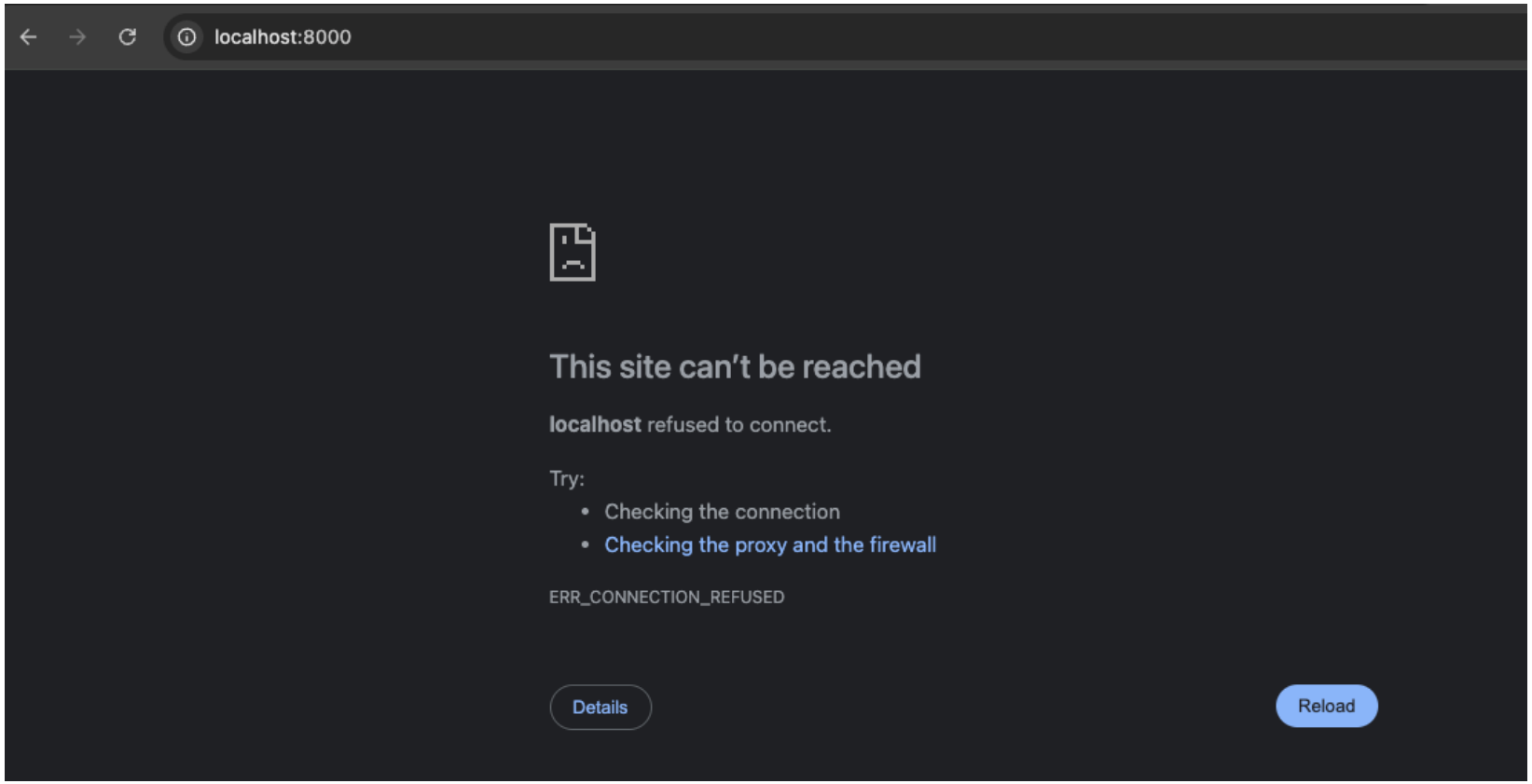
```
thuannd@MacNaN backend % docker logs --help

Usage:  docker logs [OPTIONS] CONTAINER

Fetch the logs of a container

Aliases:
  docker container logs, docker logs

Options:
      --details        Show extra details provided to logs
  -f, --follow         Follow log output
      --since string   Show logs since timestamp (e.g. "2013-01-02T13:23:37Z")
  -n, --tail string    Number of lines to show from the end of the logs (defau
  -t, --timestamps     Show timestamps
      --until string   Show logs before a timestamp (e.g. "2013-01-02T13:23:37
```

```
thuannd@MacNaN backend % docker logs backend_cont
INFO     Using path main.py
INFO     Resolved absolute path /app/main.py
INFO     Searching for package file structure from directories with __init__.py files
INFO     Importing from /

 Python package file structure

 app
 ├── __init__.py
 └── main.py

INFO     Importing module app.main
INFO     Found importable FastAPI app

 Importable FastAPI app

 from app.main import app

INFO     Using import string app.main:app

                FastAPI CLI - Production mode

 Serving at: http://0.0.0.0:8000

 API docs: http://0.0.0.0:8000/docs

 Running in production mode, for development use:

 fastapi dev

INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
thuannd@MacNaN backend %
```

Docker Build

❖ Mapping Port

```
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```



Docker Build

❖ Mapping Port

docker run -it --name backend_cont -p 8000:8000 backend:v0.0.0

```
thuannd@MacNaN backend % docker run -it --name backend_cont -p 8000:8000 backend:v0.0.0
INFO     Using path main.py
INFO     Resolved absolute path /app/main.py
INFO     Searching for package file structure from directories with __init__.py files
INFO     Importing from /

 Python package file structure

 app
 ├── __init__.py
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INFO     Importing module app.main
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 FastAPI CLI - Production mode

 Serving at: http://0.0.0.0:8000

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 Running in production mode, for development use:

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INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```
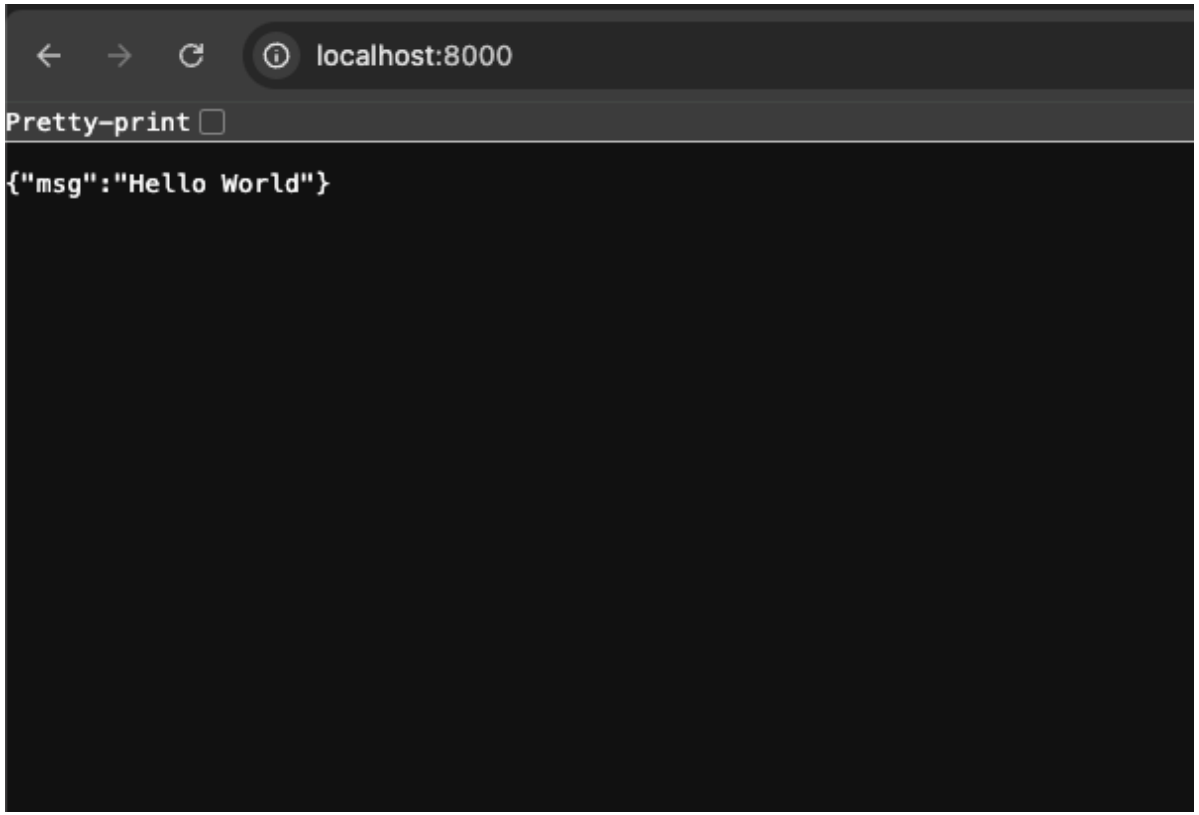
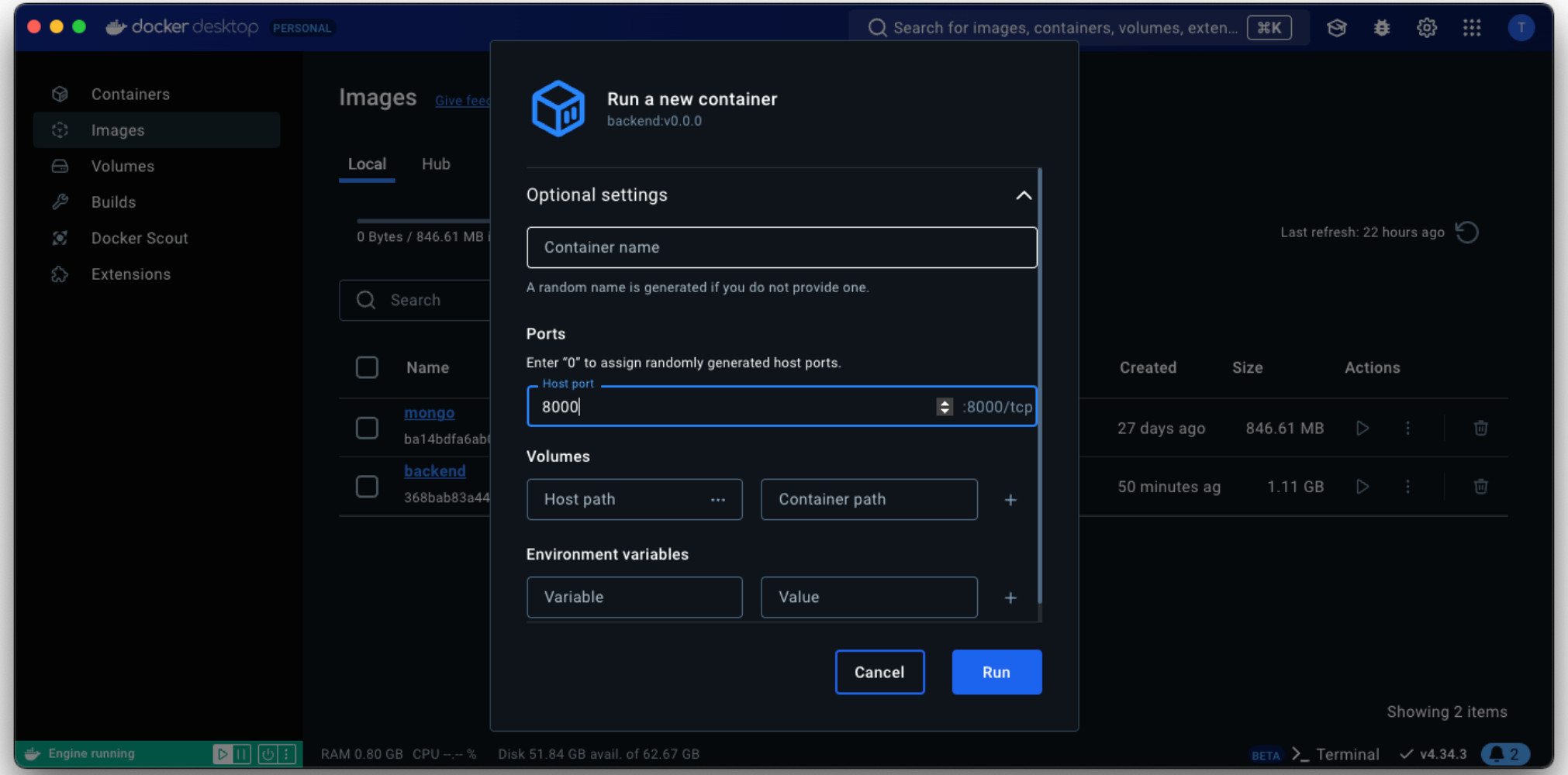



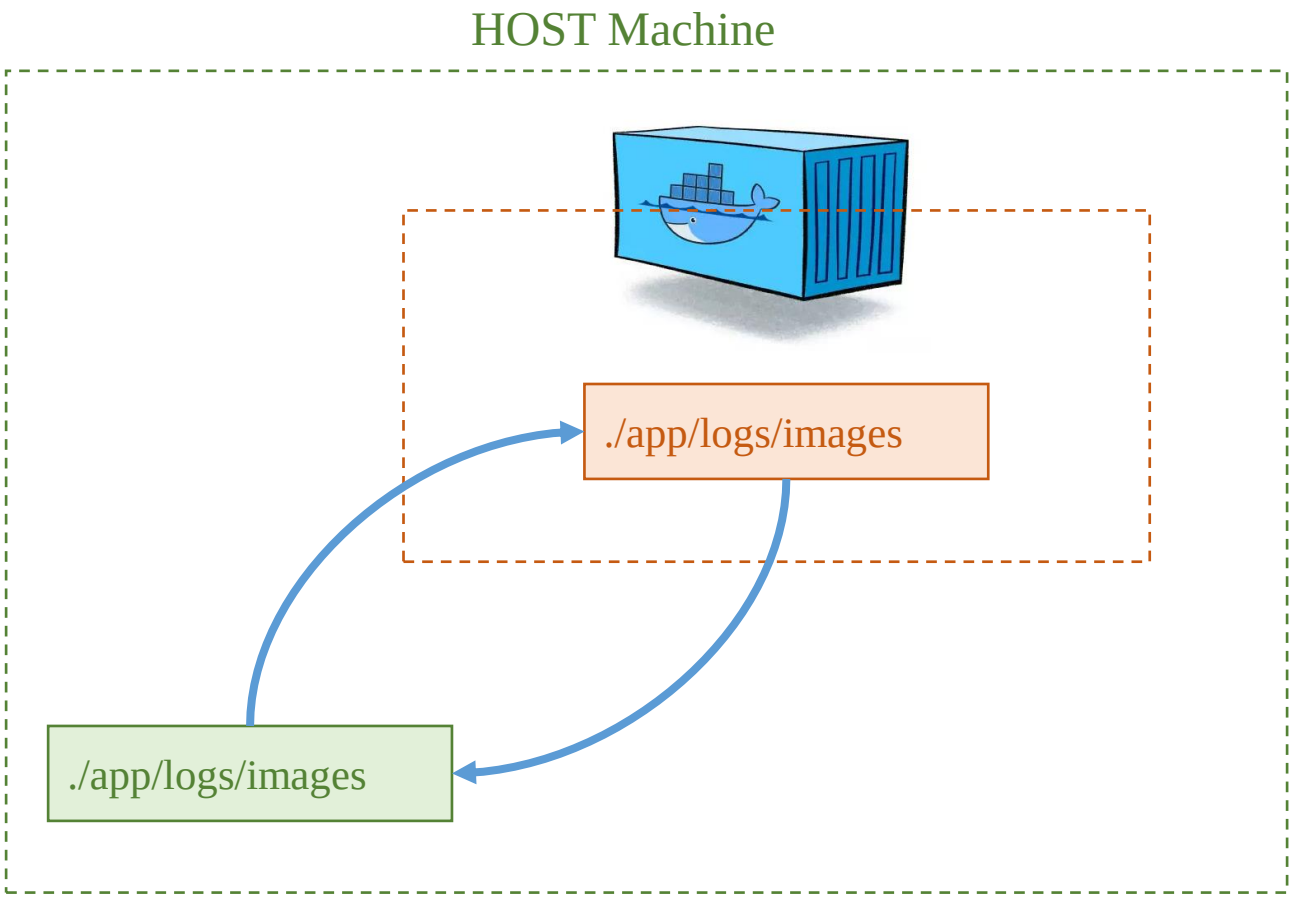
Docker Build

❖ Mapping Port



Docker Build

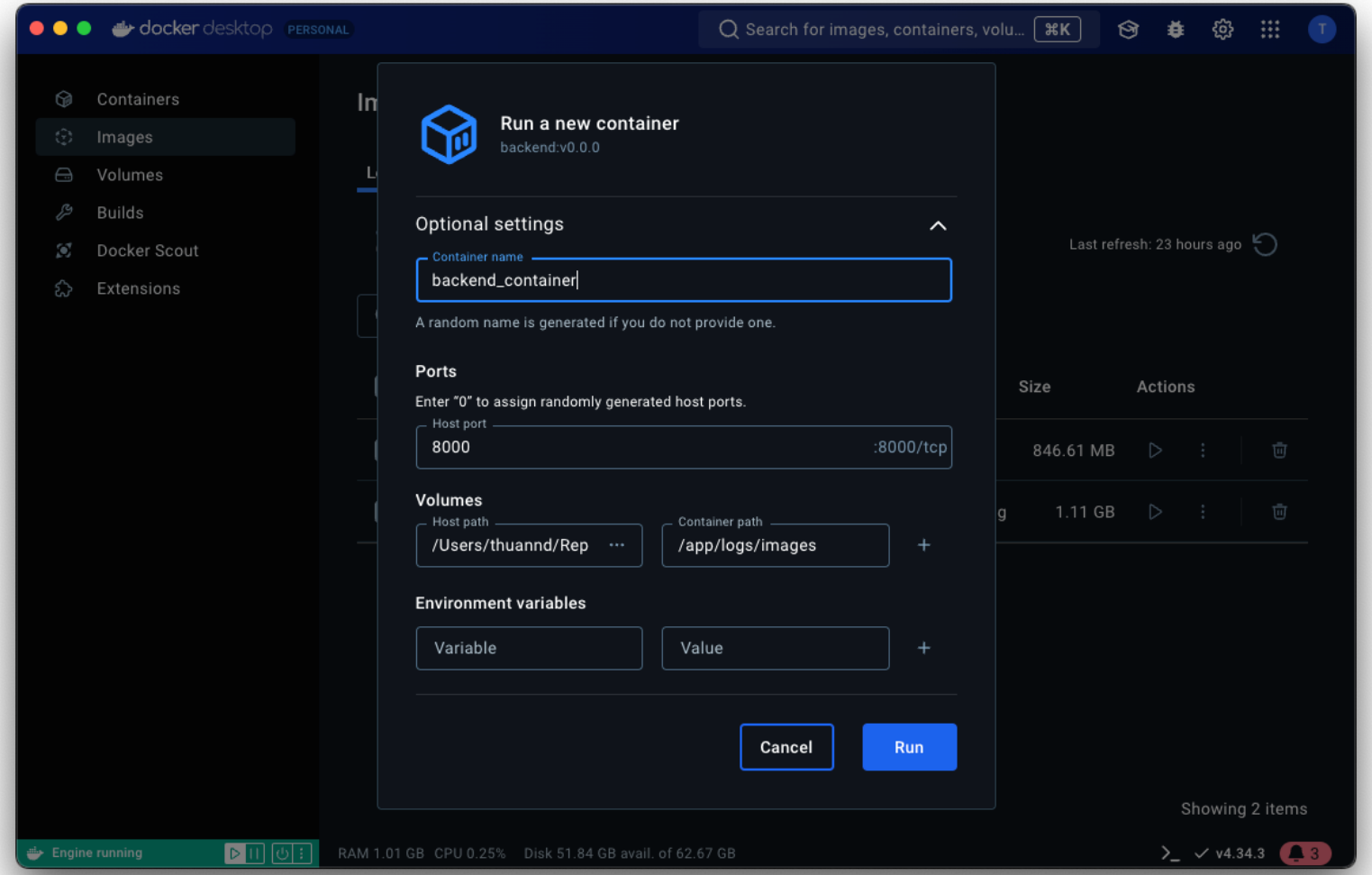
❖ Bind Mount Volume



Docker Build

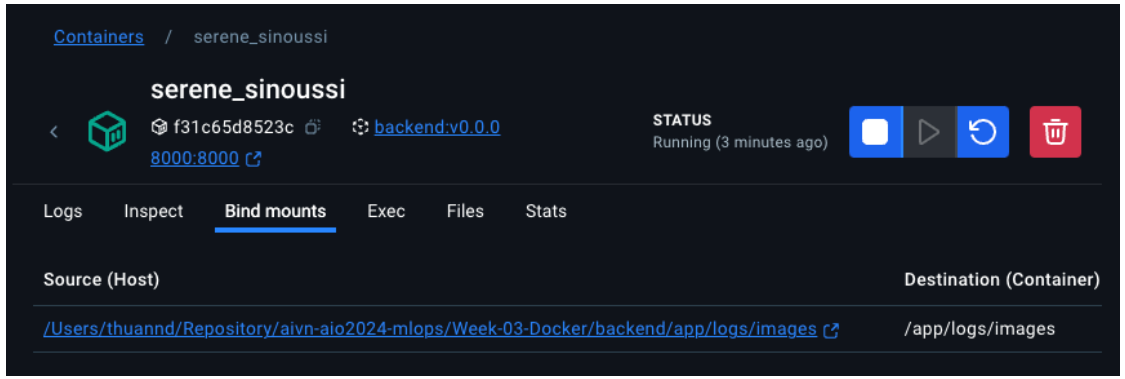
❖ Bind Mount Volume

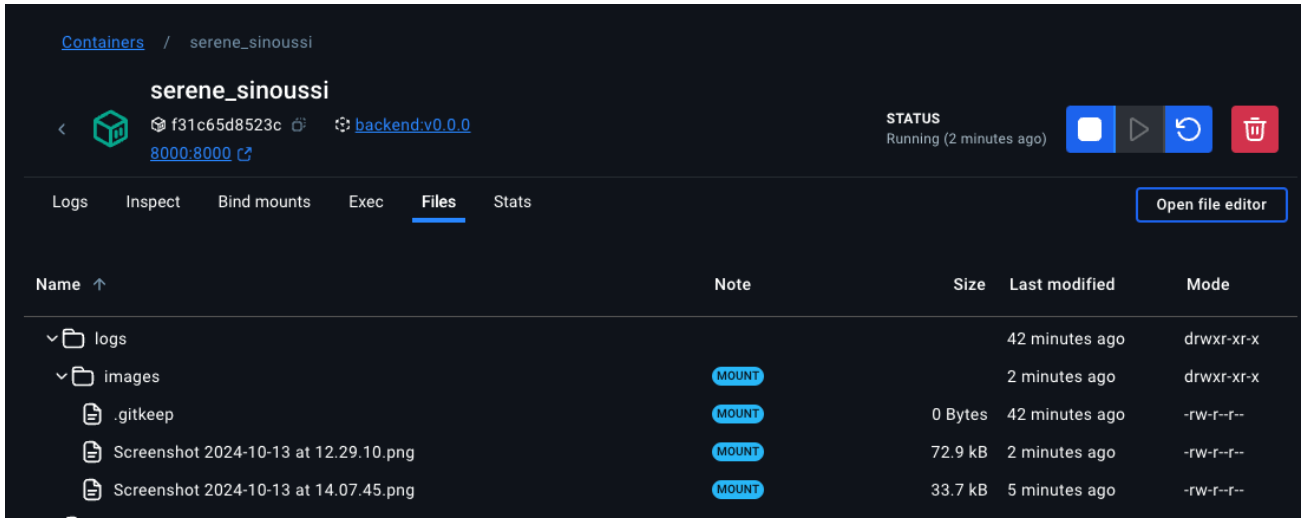
```
docker run -it --name backend_cont -p 8000:8000 -v .app/logs/images:/app/logs/images backend:v0.0.0
```

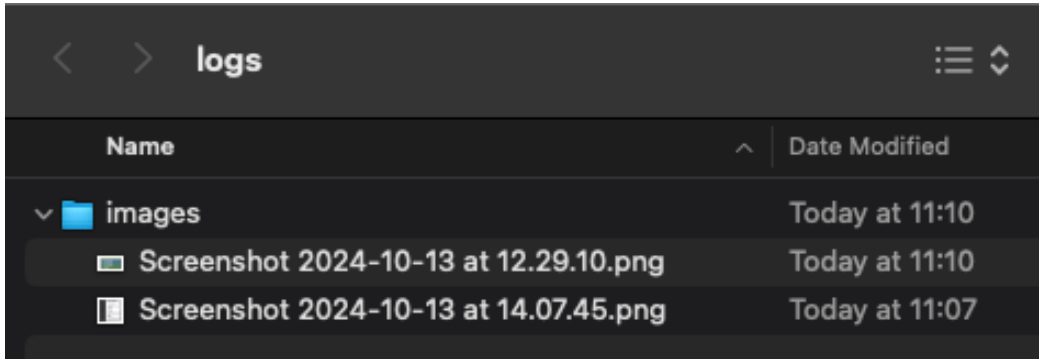


Docker Build

❖ Bind Mount Volume





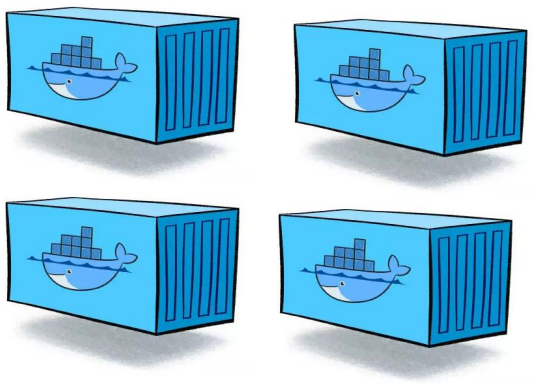


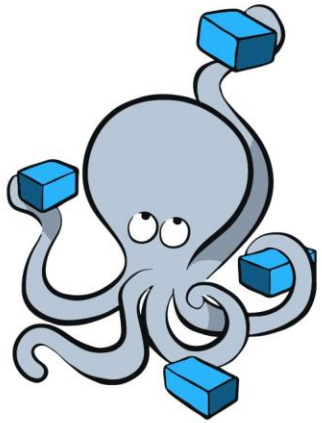
Docker Compose

Docker Compose

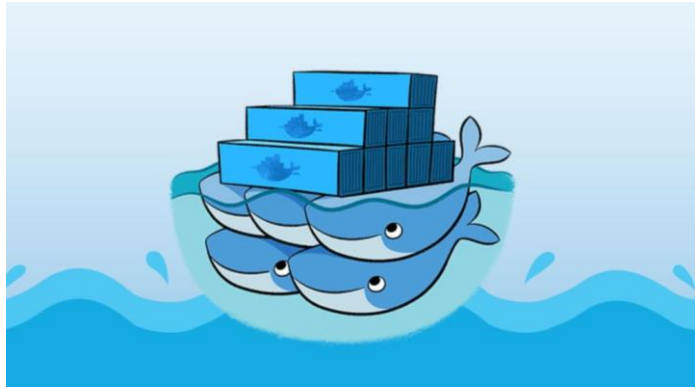
❖ Getting Started

- Docker Compose is a tool for defining and running multi-container applications.
- Allow control of your entire application stack, making it easy to manage services, networks, and volumes in a single comprehensible YAML configuration file.





docker
Compose



Ref: https://docs.docker.com/compose

Docker Compose

❖ Configuration file

docker-compose.yml

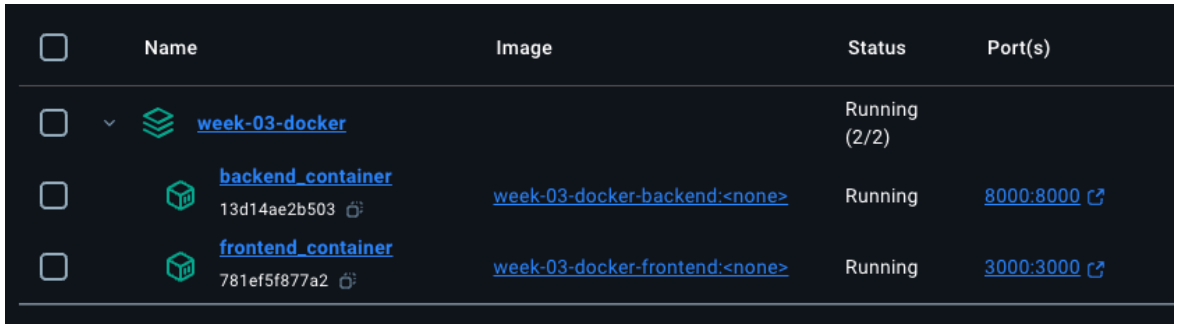
```yaml
services:
  backend:
    build:
      context: ./backend
    container_name: backend_container
    volumes:
      - ./backend/app/logs/images:/app/logs/images
    ports:
      - 8000:8000
    entrypoint: "uvicorn app.main:app --host 0.0.0.0 --port 8000"

  frontend:
    build:
      context: ./frontend
    container_name: frontend_container
    ports:
      - 3000:3000
    entrypoint: "python app.py"

```

Docker Compose

❖ Run Compose

Usage: docker compose [OPTIONS] COMMAND

docker compose up -d




```
thuannd@MacNaN Week-03-Docker % docker compose up -d
[+] Building 6.1s (20/20) FINISHED
 => [frontend internal] load build definition from Dockerfile
 => => transferring dockerfile: 270B
 => [backend internal] load build definition from Dockerfile
 => => transferring dockerfile: 333B
 => [backend internal] load metadata for docker.io/library/python:3.11.9-slim
 => [backend internal] load .dockerignore
 => => transferring context: 2B
 => [frontend internal] load .dockerignore
 => => transferring context: 2B
 => [frontend 1/5] FROM docker.io/library/python:3.11.9-slim@sha256:8fb099199b9f2d70342674bc
 => [backend internal] load build context
 => => transferring context: 2.22kB
 => [frontend internal] load build context
 => => transferring context: 1.26kB
 => CACHED [frontend 2/5] WORKDIR /app
 => CACHED [frontend 3/5] COPY ./requirements.txt /app/requirements.txt
 => CACHED [frontend 4/5] RUN pip install --no-cache-dir --upgrade -r /app/requirements.txt
 => CACHED [frontend 5/5] COPY ./app /app
 => CACHED [backend 2/5] WORKDIR /src
 => CACHED [backend 3/5] COPY ./requirements.txt /src/requirements.txt
 => CACHED [backend 4/5] RUN pip install --no-cache-dir --upgrade -r /src/requirements.txt
 => CACHED [backend 5/5] COPY ./app /src/app
 => [frontend] exporting to image
 => => exporting layers
 => => writing image sha256:1da40ef78601761ebaa94959c0b03f17cda3e7ece39becfdee365cd3f555adb5
 => => naming to docker.io/library/week-03-docker-frontend
 => [backend] exporting to image
 => => exporting layers
 => => writing image sha256:328b892c967db1ec5c6fdb40535396d076948c446944251797752c32e3f62df6
 => => naming to docker.io/library/week-03-docker-backend
 => [frontend] resolving provenance for metadata file
 => [backend] resolving provenance for metadata file
[+] Running 3/3
 ✔ Network week-03-docker_default  Created
 ✔ Container backend_container     Started
 ✔ Container frontend_container    Started
```

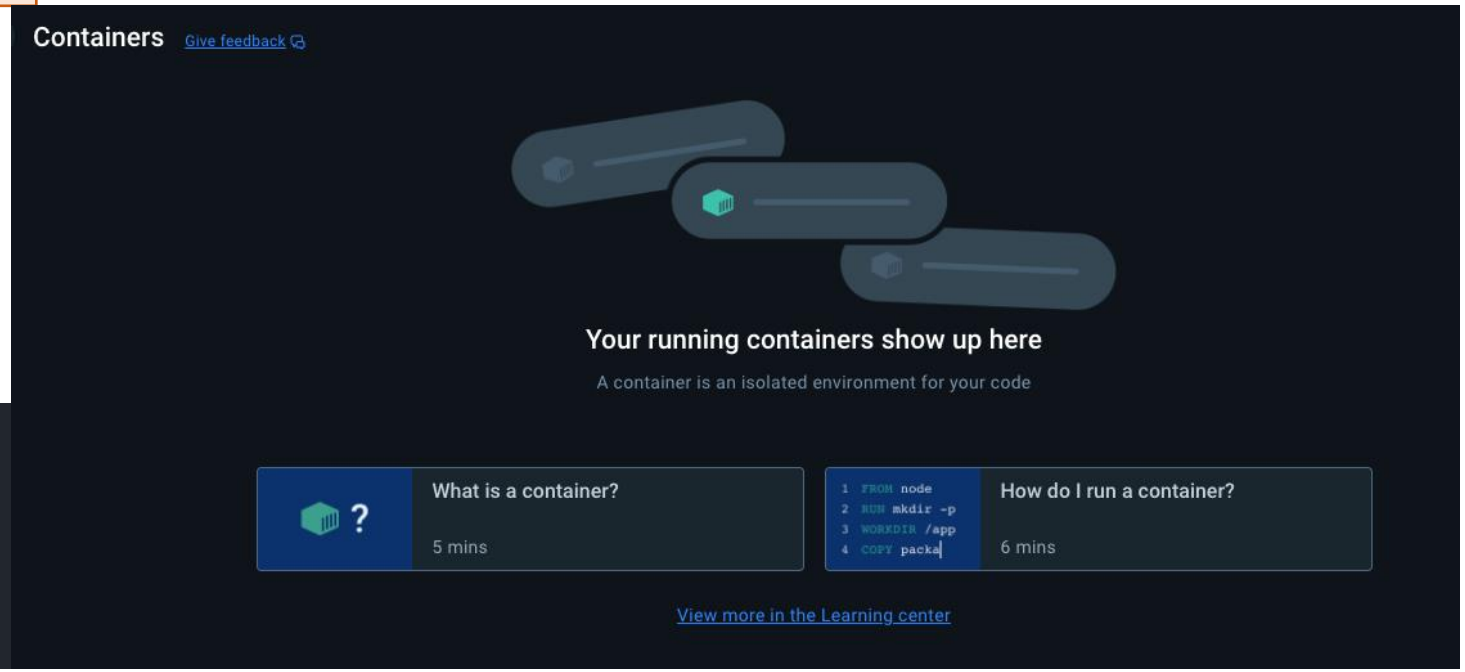
Docker Compose

❖ Stop Compose

Usage: docker compose [OPTIONS] COMMAND

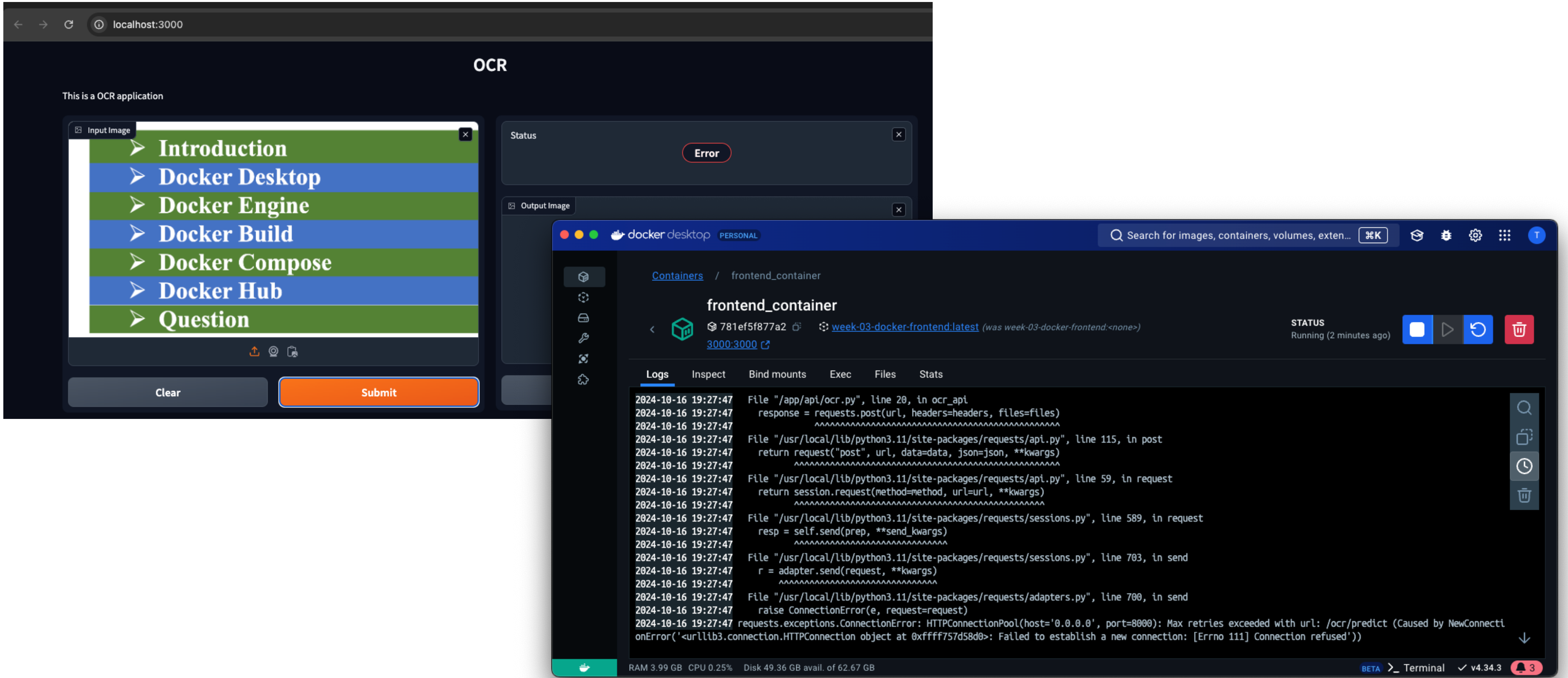
docker compose down

```
thuannd@MacNaN Week-03-Docker % docker compose down
[+] Running 3/3
 ✔ Container backend_container      Removed
 ✔ Container frontend_container     Removed
 ✔ Network week-03-docker_default   Removed
thuannd@MacNaN Week-03-Docker %
```



Docker Compose

❖ Docker Compose Network



Docker Compose

❖ Docker Compose Network

Config network in docker-compose.yml file

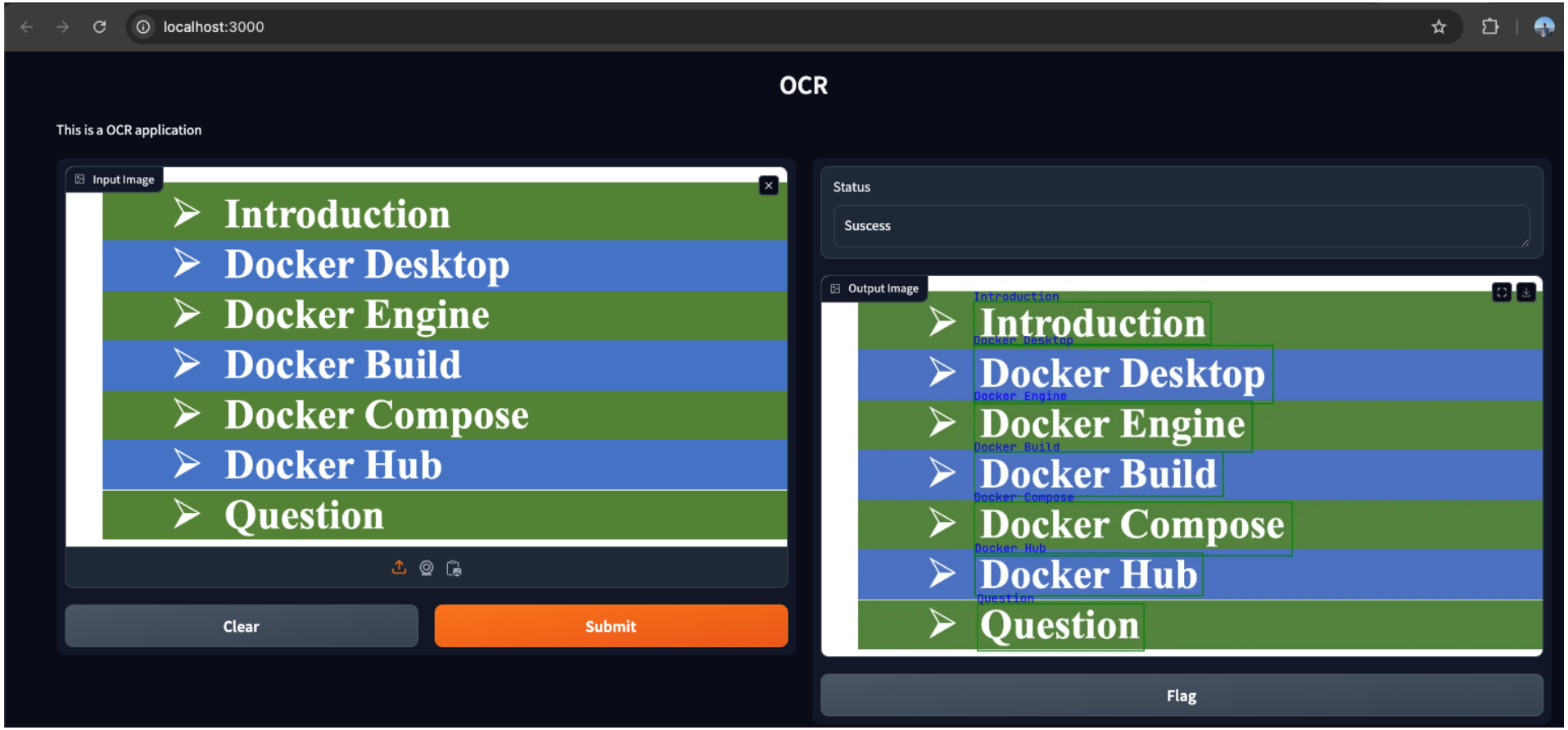
docker compose up -d

```
 => [backend] resolving provenance for metadata fil
[+] Running 3/3
 ✔ Network week-03-docker_aio_mlops_net  Created
 ✔ Container backend_cont                Started
 ✔ Container frontend_cont               Started
```

```yaml
services:
  backend:
    build:
      context: ./backend
    container_name: backend_cont
    volumes:
      - ./backend/app/logs/images:/app/logs/images
    networks:
      - aio_mlops_net
    ports:
      - 8000:8000
    entrypoint: "uvicorn app.main:app --host 0.0.0.0 --port 8000"

  frontend:
    build:
      context: ./frontend
    container_name: frontend_cont
    env_file:
      - ./frontend/.env
    networks:
      - aio_mlops_net
    ports:
      - 3000:3000
    entrypoint: "python app.py"

networks:
  aio_mlops_net:
    driver: bridge
```

Docker Compose

❖ Docker Compose Network

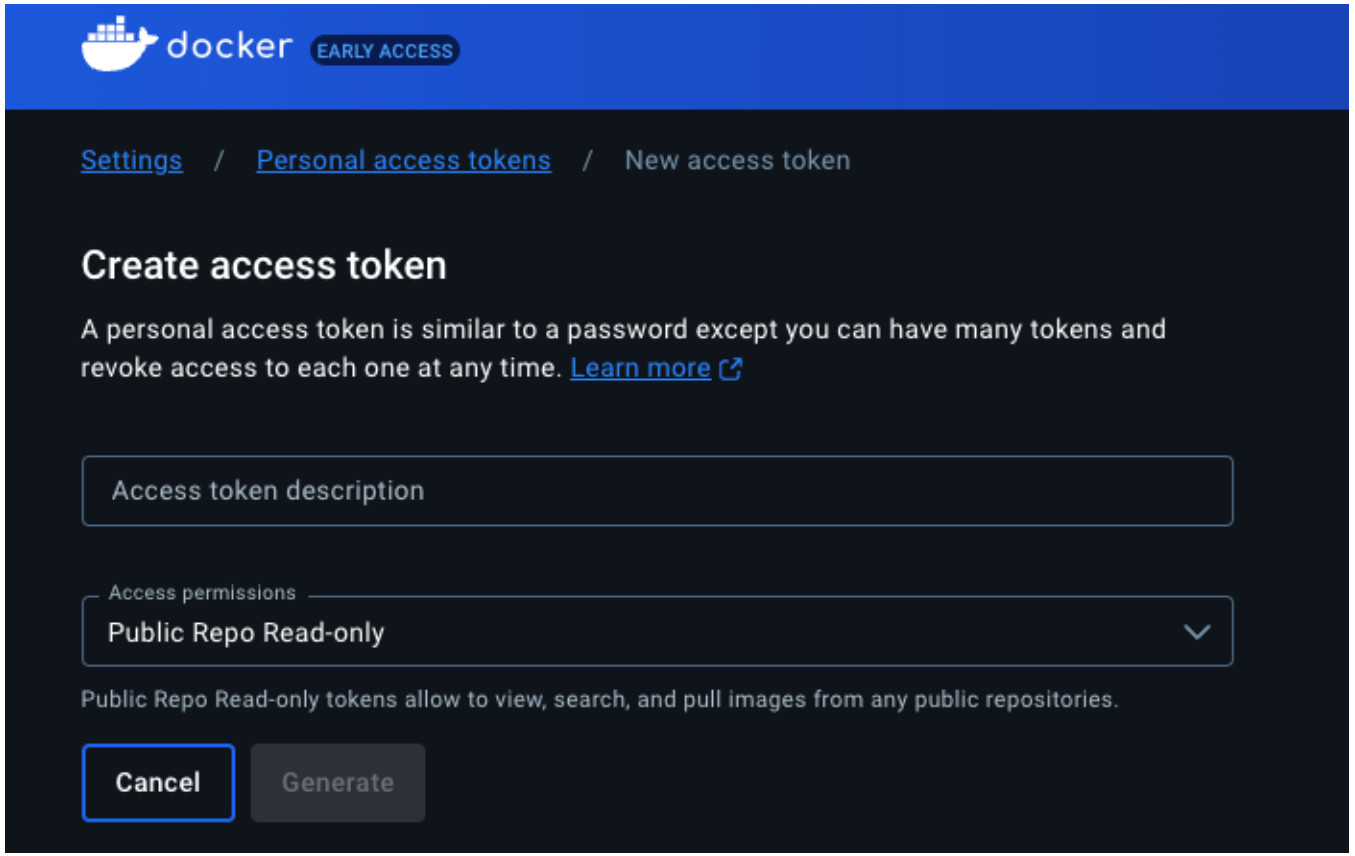


Docker Hub

Docker Hub

❖ Getting Started

- Login by token at: https://app.docker.com/settings/personal-access-tokens
- Tagging your local image with Docker Hub username.
- Push image to Hub

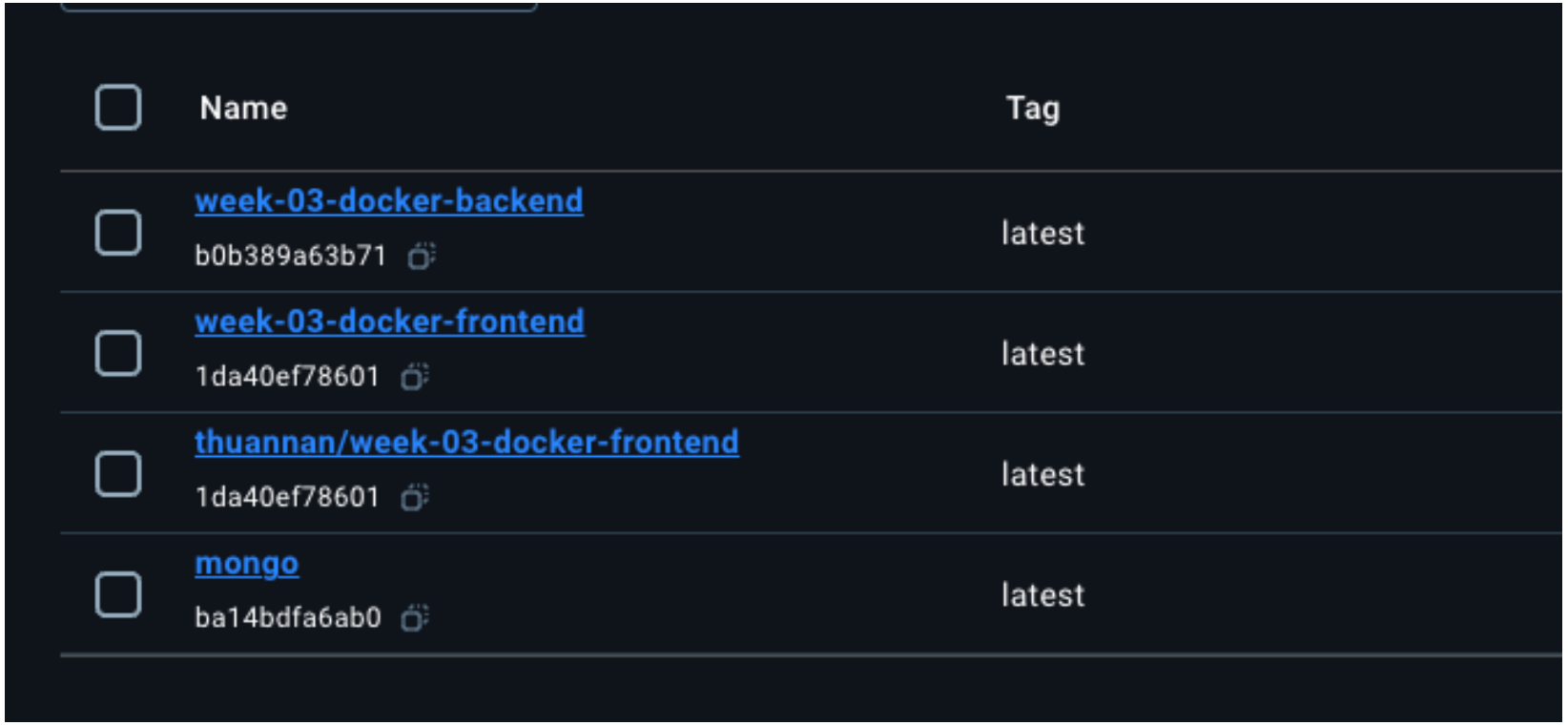


Docker Hub

❖ Tagging

- Login by token at: https://app.docker.com/settings/personal-access-tokens
- Tagging your local image with Docker Hub username.
- Push image to Hub

```
docker tag week-03-docker-frontend thuannan/week-03-docker-frontend
```



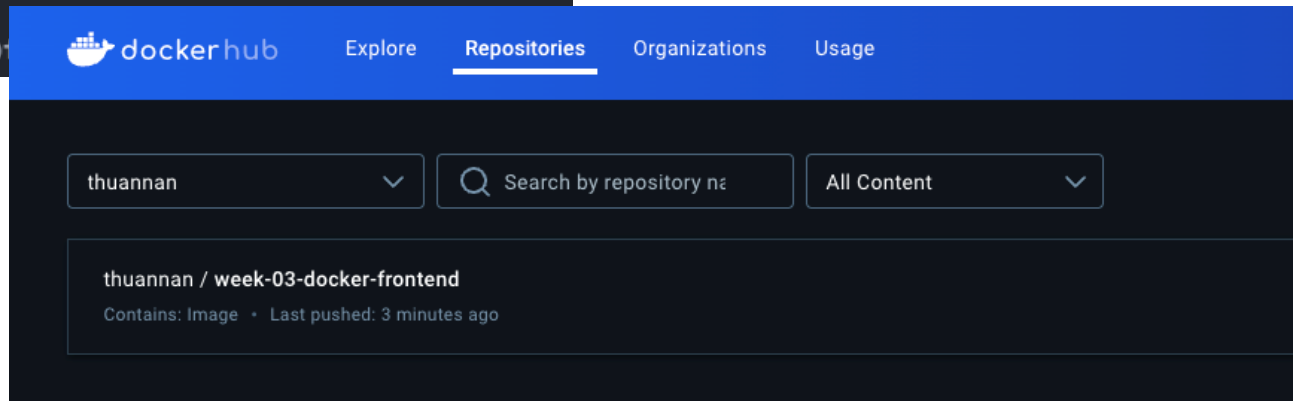
Docker Hub

❖ Pushing

- Login by token at: https://app.docker.com/settings/personal-access-tokens
- Tagging your local image with Docker Hub username.
- Push image to Hub

docker push thuannan/week-03-docker-frontend

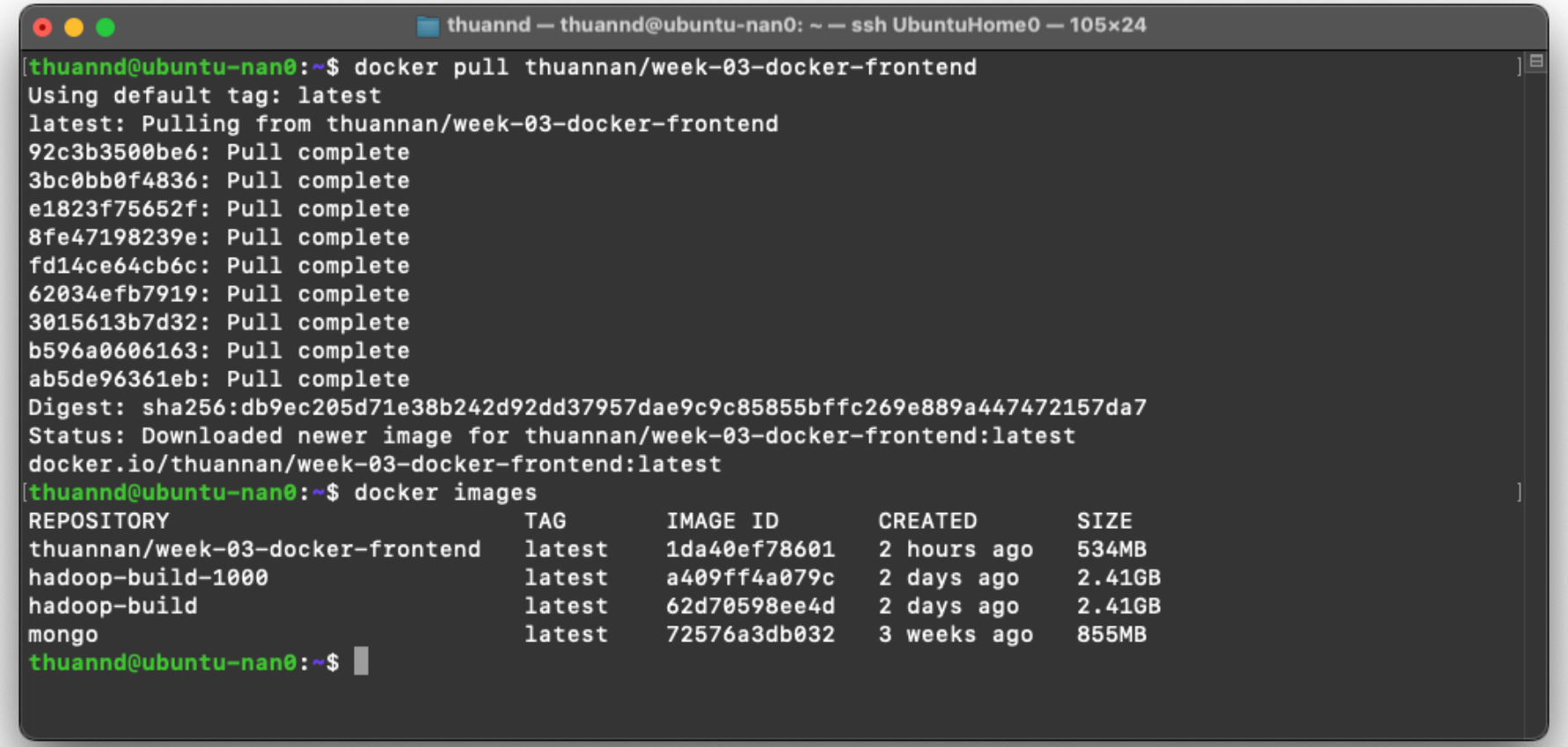
```
(aio-mlops-w2) thuannd@MacNaN Week-03-Docker % docker push thuannan/week-03-docker-frontend
Using default tag: latest
The push refers to repository [docker.io/thuannan/week-03-docker-frontend]
0723cc98afde: Pushed
5aa879153ec6: Pushed
c5505ccb471b: Pushed
5c143465292d: Pushed
74c21085cca9: Pushed
84e14a567d37: Pushed
9137ed4faf20: Pushed
ad0a49486599: Pushed
e644ff0c302d: Pushed
latest: digest: sha256:db9ec205d71e38b242d92dd37957dae9c9c85855b
```



Docker Hub

❖ Pulling

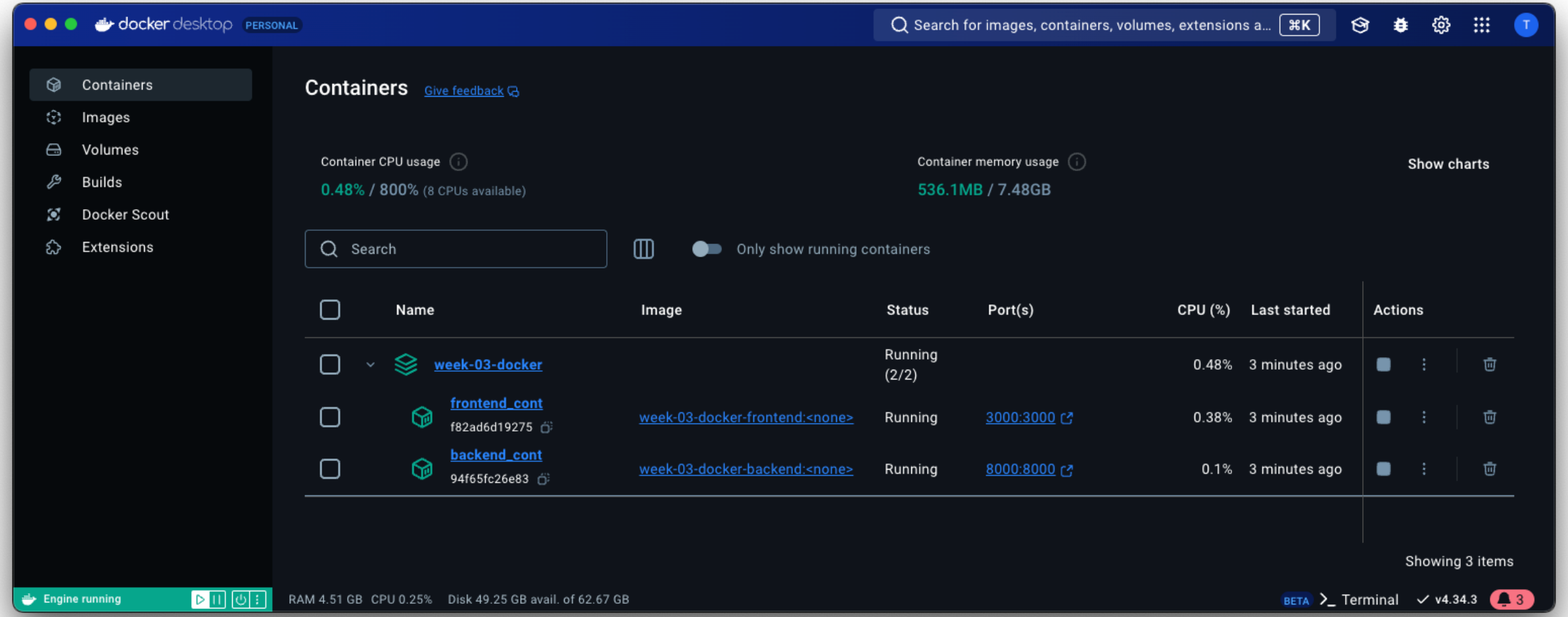
docker push thuannan/week-03-docker-frontend



Summary

Summary

❖ Performance



Summary

❖ Docker CLI: image

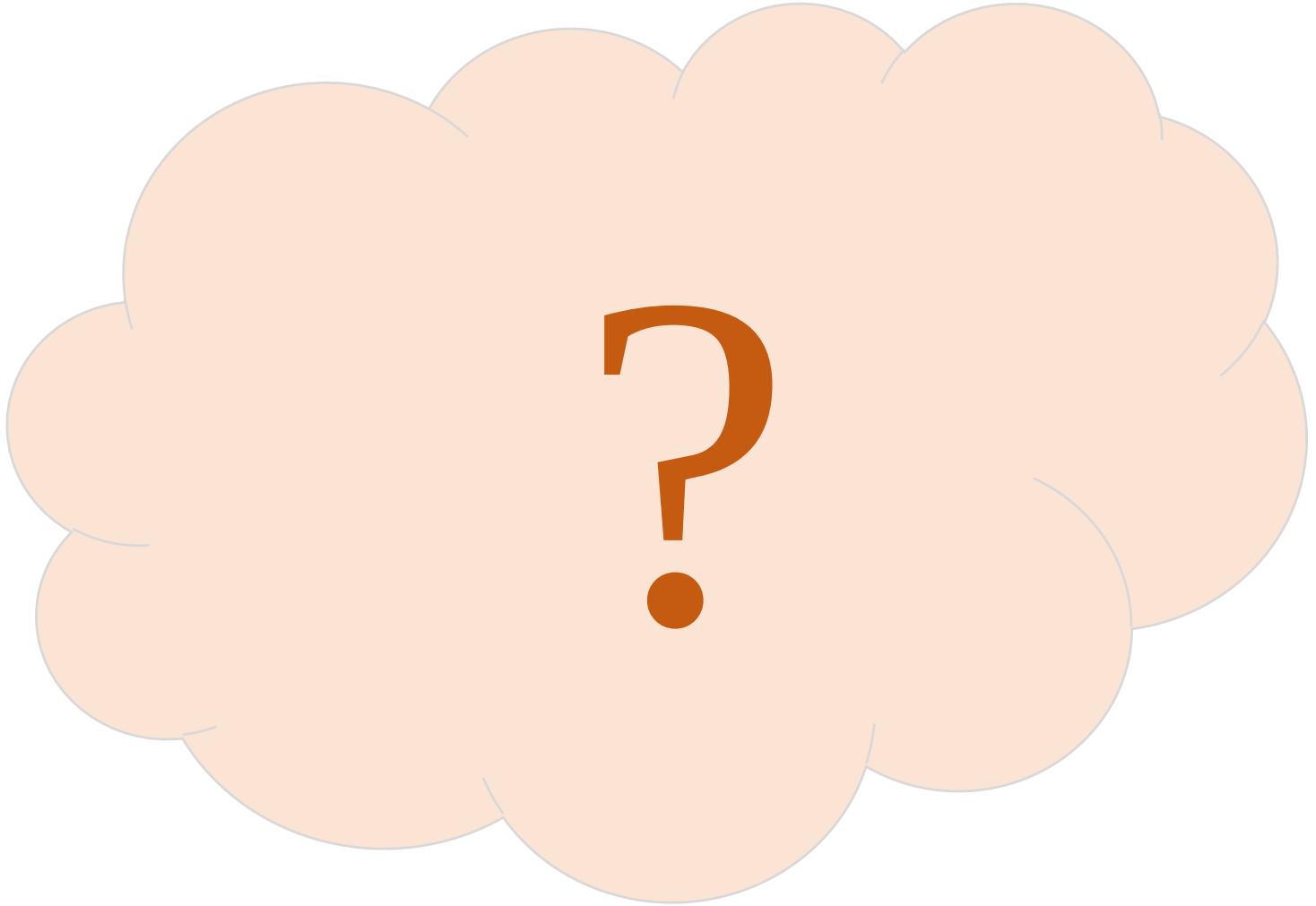
Command	Function
docker images	To list all the Docker images on your system
docker pull <image_name>	To download a Docker image from registry
docker build -t <image_name>:<tag> <path_to_dockerfile>	To create a new image from a Dockerfile
docker rmi <image_name_or_id>	To remove a Docker image
docker tag <existing_image> <new_image_name>:<tag>	To tag an image with a new name
docker push <image_name>:<tag>	To push an image to a Docker registry

Summary

❖ Docker CLI: container

Command	Function
docker ps	To list all running containers
docker run <image_name>	To create and run a new container from an image
docker stop <container_id_or_name>	To stop a running container
docker start <container_id_or_name>	To start a container that was previously stopped
docker rm <container_id_or_name>	To delete a container
docker logs <container_id_or_name>	To see the logs of a container
docker exec -it <container_id_or_name> <command>	To run a command inside an already running container

Question




Thank you!